

## Review

# An Up-to-Date Review of Passive Building Envelope Technologies for Sustainable Design

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**Abstract:** A primary driving force of today's urban environment is the development or enhancement of building stock with a focus on minimizing its environmental footprint, eliminating its dependence on fossil fuels, enforcing its energy efficiency and self-sufficiency, and helping alleviate climate change. Therefore, in the present study, an up-to-date review regarding the passive building retrofitting techniques for sustainable and energy efficiency design is conducted. Numerous passive building solutions and design concepts are thoroughly examined in terms of innovation and energy-saving potential. The examined techniques include novel thermal insulation materials, innovative windows systems, high thermal mass technologies, optically advanced coatings appropriate for cooling abatement, and various energy-efficient bioclimatic designs, for instance, shading techniques, mechanical ventilation in combination with heat recovery, and green roofs and façades. The scope of the present review is to thoroughly and comparatively investigate passive building energy retrofit solutions as presented in the recent scientific literature mainly within the last five or up to ten years. The passive, energy-mitigating solutions are examined in terms of energy savings primarily in residential buildings, but also in tertiary buildings, as well as of specific investment costs. Lastly, an extensive discussion evaluating the comparative advantages and disadvantages of the examined passive envelope technologies is conducted, allowing a comprehensive and multilevel comparison.

**Keywords:** passive retrofitting; energy savings; building renovation; energy efficiency; thermal performance enhancement



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## 1. Introduction

### 1.1. Identification of the Problem

Climate change and global warming have emerged as one of the most significant threats to our planet. Human-induced warming reached around 1 °C above pre-industrial levels in 2017, increasing by approximately 0.2 °C per decade [1]. Predictions about future weather conditions agree that the global surface temperature will increase by 1.5 °C and 2.0 °C during the 21st century unless radical reductions in emissions of CO<sub>2</sub> and other greenhouse gases take place in the coming decades [2]. Temperatures higher than the global average have been recorded in many regions and seasons, with 20% to 40% of the global human population having already experienced at least a seasonal warming of more than 1.5 °C during the decade between 2006 and 2015 [1]. Aside from the changes in the mean climate, an increase in the frequency and amplitude of extreme weather events has also been observed. Although hot extremes belong to the natural climate variability, global warming is responsible for 75% of the daily hot extremes [3]. Global warming and urban heat islands are two interdependent phenomena that lead to an increase in a cities' temperature and the exacerbation of the energy cooling demand of the building stock [4]. Around half a million cities have presented symptoms of the urban heat island phenomenon [5], deteriorating their citizens' quality

of life, causing them health problems [6], and intensifying energy poverty discretization. The high density of buildings, the usage of materials of high heat absorptance on the building envelope [7] of the city's public constructions, and the minimization of tree-planted areas combined with the heat generated by human activities are related to the urban heat island phenomenon and an average increase in the buildings' cooling load up to 13% for the case of city centers compared with rural areas [6]. The increase in population, technological advancements, and prosperity, combined with income growth have led and will inevitably continue to lead to urban sprawl. It is predicted that the global consumption for residential cooling will escalate to 35% in 2050 and to 61% in 2100 [8].

### 1.2. Energy Use in the Building Sector

On a worldwide scale, buildings are responsible for an energy consumption of 4.19 TWh, which corresponds to 36% of the total final energy consumption. The operation of buildings accounts for 30% of the global energy consumption, while the construction procedures for the respective domain for 6% [9]. At a European level, the building sector accounts for a higher share of the final energy consumption, specifically 40%, and residential buildings are responsible for half of this [10]. It needs to be clarified that final energy consumption is defined as the total energy received and consumed by end users, such as households, industry, and agriculture. In other words, it is the quantity of user-available energy, after excluding the conversion, transmission, and distribution losses. The increasing deployment of building energy systems, and especially of space cooling energy systems, is not evenly distributed around the world but mainly observed in developed and developing economies. This geographical discrepancy explains the increased share of buildings' energy consumption for the continent of Europe.

Less than 25% of Europe's building potential is characterized by sufficient energy performance [11], without displaying any hazardous deficiencies that can lead to inadequately healthy indoor living conditions. The European ambition is to become the first climate-neutral continent by 2050 [11]. However, the increasing floor area and population, the escalating request for energy services, and the restricted enhancements in energy efficiency inevitably lead to the rise in the demands of the building sector [12]. Figure 1 illustrates the global building floor area for advanced and developing economies, as well as the mean energy intensity of buildings from 2010 to 2022 [10]. In particular, the global building floor area for 2022 has increased by 31.3% in comparison to 2010 [10]. The rising tendency of building floor area, the rapid decrease in occupancy rates, and the increased ownership of electrical appliances in the advanced and emerging national economies are an immediate result of the reported mean income rise [13]. These social phenomena in combination with the escalation of severe hot weather phenomena result in the soaring of building energy demands. For instance, in 2021, which was one of the warmest years ever recorded globally, the highest annual growth of space cooling demand among all buildings' end use was recorded, representing 16% of the building sector's final electricity consumption (about 2000 TWh) [10].

The residential sector is responsible for almost 70% of the final energy consumption in the building sector [13]. Within the European Union (EU), it ranks as the third most significant sector in terms of energy demand, accounting for 27.9% of the final energy consumption [14]. For eight out of the twenty-seven EU member states, the residential sector is characterized as the largest national energy consumer. For these member states, the energy consumption share of residencies varies from 29.3%, for the case of Germany, to 35.1% for the case of Croatia [15]. In Greece, residential energy consumption accounts for 28.8% of the nation's total final energy use [15]. Heating, cooling, and domestic hot water (DHW) preparation make up approximately 79% of households' energy usage in buildings [16]. Specifically, in the EU, the most energy-demanding type of end use is space heating, with a share of 64.4% of the final energy consumption [16]. The corresponding value for Greece is 53.2%. The second largest energy-demand type of end use is water heating, with a share of 14.5% for the EU, and 14.0% for Greece. Space cooling accounts for

only 0.5% of the final energy consumption for the EU [16]. For southern European climates, space cooling demand accounts for a greater share of the final energy consumption in buildings. Specifically, the corresponding share for Greece is equal to 4.1%, for Malta it is 17.0%, for Cyprus it is 10.6%, for Albania it is 7.95%, and for Kosovo it is 4.96% [16]. Space cooling is the fastest-growing end use with a growth rate of around 33% between 2000 and 2018, and 5% between 2017 and 2018 [10]. The annual growth of air conditioner sales for 2022 globally and in Europe was reported to equal 37.0% and 19.0%, respectively [17]. Although the gradually increasing penetration of space cooling systems results in the rise in energy consumption for cooling, it also results in the enhancement of thermal comfort conditions and indoor quality of life. Figure 2 depicts the share of the final energy consumption of every type of end use regarding residential buildings globally, for the European Union, and for Greece.

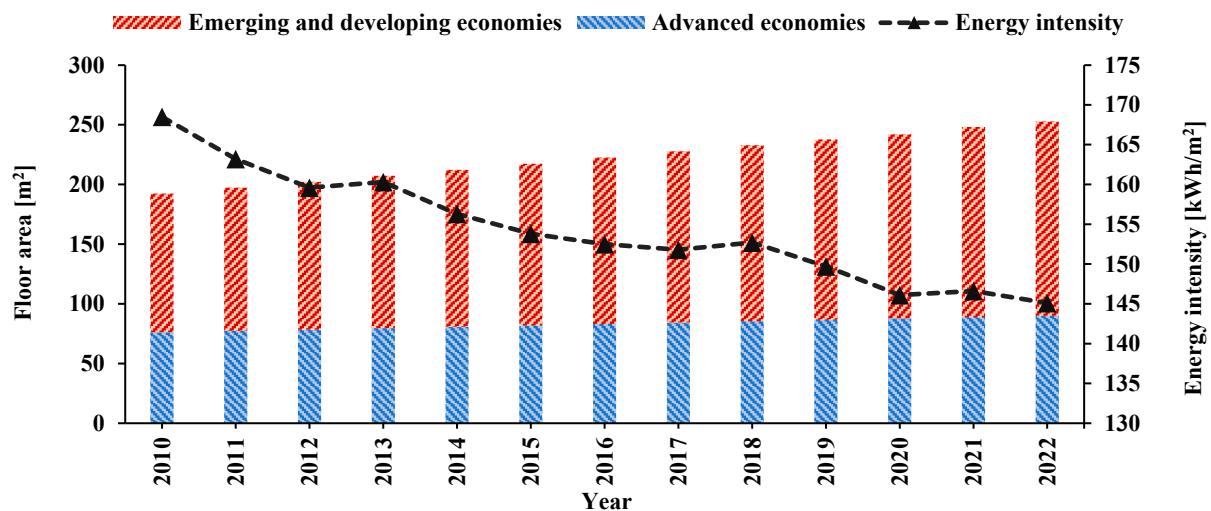


Figure 1. Global floor area and buildings' energy intensity from 2010 to 2023 (data were retrieved from Ref. [10]).

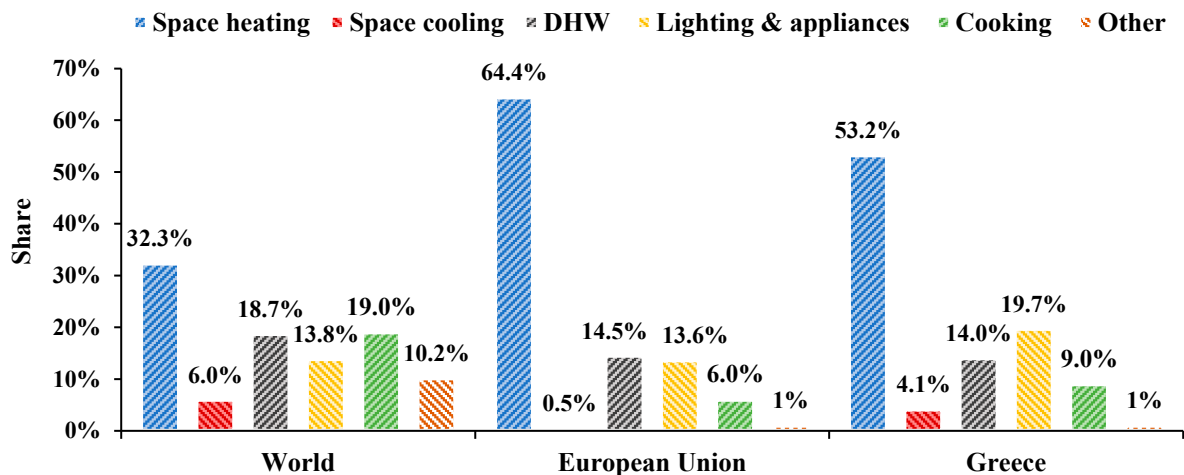


Figure 2. Share of final energy consumption by type of end use (data were retrieved from Ref. [10]).

### 1.3. Thermal Comfort and Energy Poverty and Heat-Related Mortality

The rising tendency of energy prices and the inflationary pressures in the market of consumer goods and services constantly undermine the financial prosperity of households, setting them under threat of energy poverty [18]. Energy poverty in households is defined as the inability to access essential energy services that support a decent standard of living and health, such as adequate warmth through heating, and cooling, as temperatures rise, lighting, and energy to power appliances [19]. According to the European Commission [20],

the phenomenon of energy poverty stems from a combination of low income, a large share of available income allocated to energy expenses, and the low energy efficiency of buildings. Regarding the price of energy, in the first semester of 2023, compared to the same semester of the previous year, the price of electricity for household consumption increased by 12.6% for the European Union and 9.6% for Greece [21]. Similarly, the price of natural gas for household consumption increased by 34.8% for the EU and 14.2% for Greece [21]. In 2022, around 41.5 million people in the EU were unable to afford to keep their homes sufficiently warm [21]. That number corresponds to 9.3% of the households in the European Union, while for Greece, the share is even higher and equal to 18.7% [16].

Energy poverty is a crucial socio-economic phenomenon that mainly concerns the countries in Central and South-Eastern Europe and is immediately connected to various serious implications for health, well-being, and social inclusion [20]. Additionally, in combination with cold, and especially, heat extremes, it poses a severe threat to human health. It is important to highlight that during the summer of 2022, more than 61,000 deaths among the European population were reported as heat-related [22]. The protagonists of this horrific statistic were countries in southern Europe, specifically Italy, Greece, Spain, and Portugal, with heat mortality rates of 295, 280, 237, and 211 deaths per million, respectively. Significant mortality remains associated with cold extremes, posing a serious health threat. Even though cold extremes are anticipated to decrease in Europe, especially in southern regions, the projected decline is not expected to compensate for the additional heat-related deaths [22].

#### *1.4. Energy Efficiency in the Building Sector*

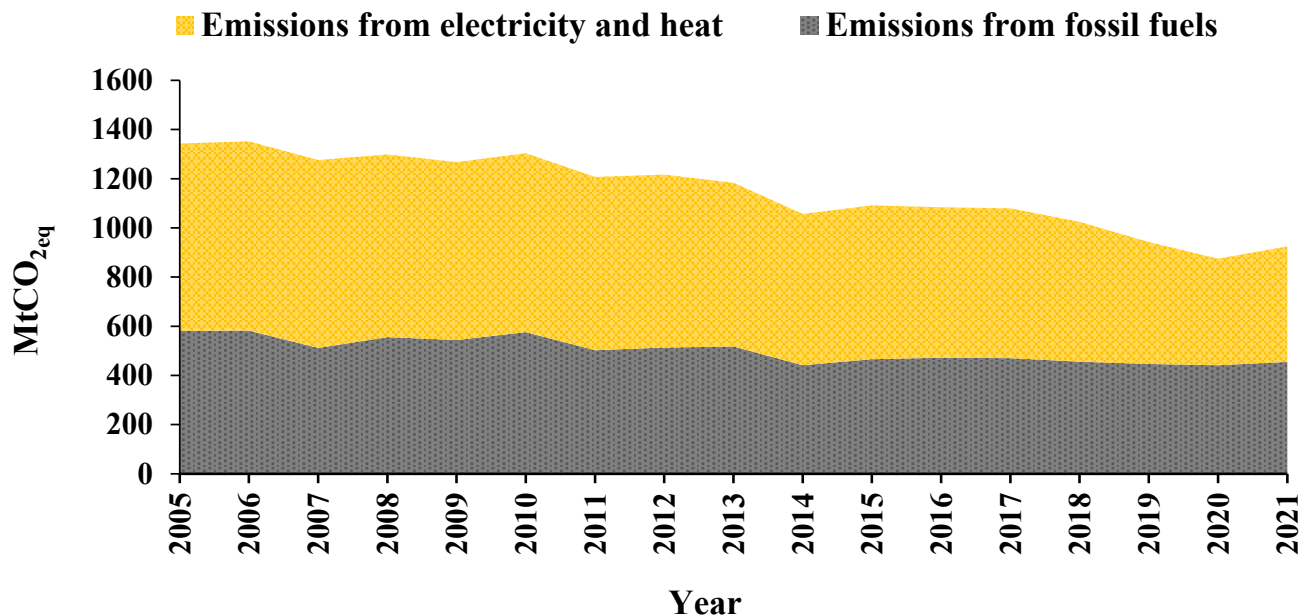
In line with the sustainable development goals for buildings and infrastructure, the net-zero-emissions and zero-carbon-ready buildings by the 2050 scenario highlight the importance of accelerating building decarbonization [23] beyond the concept of nearly zero-energy (nZEBs) and zero-energy buildings (ZEBs). This involves implementing more energy-efficient interventions [24] and utilizing electricity or district heating as energy supplies. The EU highly acknowledges this energy policy momentum through the enactment of numerous innovative directives and energy standards [9]. The 2023 adopted revision of the energy performance of building directive defines stricter and higher minimum energy performance standards for the existing public and residential building stock, timelines to achieve them, and shorter deadlines for the full transformation to zero-emissions buildings. The enhancement of energy efficiency in buildings is in alignment with the clean energy transition and the rapid independence from fossil fuels [25].

European policy and funding resources have inarguably augmented the establishment of high energy efficiency in new buildings. Specifically, new constructions consume only half the energy of constructions built over 20 years ago [21]. However, 85% of buildings in the EU were built over 20 years ago, and 85–95% are expected to still be standing in 2050 [21]. Therefore, there is a great necessity for deep energy building retrofit and transformation of the European building stock to an environmentally neutral and uniform level of energy performance. By 2020, only 5% of new building construction was zero-carbon-ready around the world [10]. In addition, in Europe, the average annual energy renovation rate is low, around 1%, and as far as deep renovations and the reduction in energy use by at least 60% are concerned, the rate is even lower [16]. Additionally, the broad spectrum of social, economic, and mostly climatological differences across its geographical territories creates a two-speed Europe in the race for a net-zero energy-building sector, with southern European countries falling behind [26].

The ongoing energy transition due to the intense climate and energy crisis is based on the rapid diffusion and adoption of sustainable energy technologies and initiatives in every field of our society. The buildings sector is one of them, able to contribute significantly to the total achievable energy savings through the interventions and application of energy-saving technologies [27]. At a worldwide scale, in 2021, direct emissions from space heating rose by 5.5%, resulting in 2.5 Gt of CO<sub>2</sub> emissions, representing 80% of direct CO<sub>2</sub> emissions



in the buildings [10]. Figure 3 illustrates the greenhouse gas emissions from energy use, electricity, and heat or fossil fuels, in the building sector in Europe, according to which, in 2021, the building sector accounted for 27% of total energy sector emissions, while 19% were indirect emissions from the production of electricity and heat used in buildings [9]. From 2005 until 2021, greenhouse gas emissions from energy use in buildings were reduced by 22.0% for the EU27 and by 54.0% for Greece [9].



**Figure 3.** Greenhouse gas emissions from energy use in buildings in Europe for 2005–2021 (data were retrieved from Ref. [9]).

### 1.5. Scope of the Present Study

In the present study, an up-to-date review of the available and energy-efficient passive building technologies is conducted. The reviewed technologies concern novel thermal insulation materials, innovative windows systems, high thermal mass technologies, optically advanced coating appropriate for cooling abatement, and various energy-efficient bioclimatic designs, for instance, shading techniques, mechanical ventilation in combination with heat recovery, and green roofs and façades. The extended review primarily examines the impact of the respective techniques on thermal performance and energy savings, and secondarily on indoor thermal conditions. The literature references regarding technological applications are drawn from recent research, mainly within the last five to ten years. The present review study aims to identify the recent technological advancements in the passive building energy retrofit domain and to investigate the possible energy savings and thermal load alleviation potential in buildings. Residential buildings are the main focus of the study; however, tertiary buildings are also examined for causes of completeness. Lastly, a comprehensive comparison among the investigated energy retrofit techniques highlights the comparative advantages and disadvantages of the studied techniques in terms of both energy efficiency and investment cost.

## 2. Insulation Materials

Building insulation is a vital passive envelope solution that results in important energy savings and more stable indoor thermal conditions. In case of both deeply retrofitted and newly constructed buildings, thermal insulation is an integral action of any energy-efficient strategy for the foundation of a sustainable and energy-efficient building. This section presents a brief discussion of the most commercially popular conventional and novel insulating materials. Additionally, these materials are compared based on techno-economic criteria to highlight the most cost-effective and thermally resistant insulating material.

### 2.1. Conventional Insulation Materials

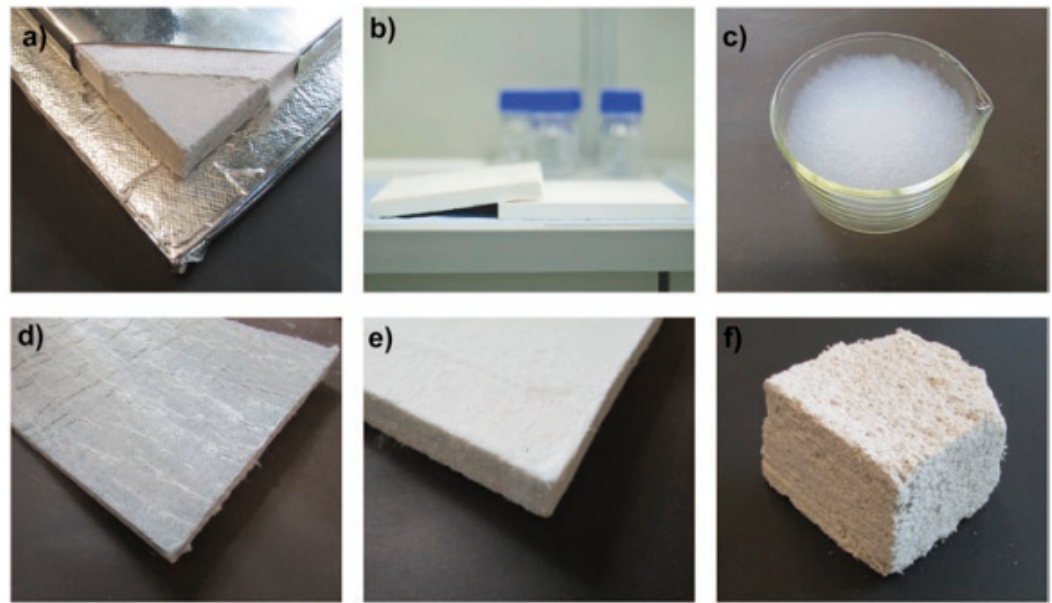
Conventional building insulation materials can be divided into two main categories: organic and inorganic insulation materials. These materials are either homogenous or composite. The organic materials include fibrous cellulose, wood fibers and cork, foamed rubber, polyisocyanurate, expanded or extruded polystyrene, polyethylene, polyurethane, phenolic foam, and other polymers. Conversely, inorganic materials include fibrous glass wool, and stone wool, as well as expanded perlite, expanded vermiculite, and ceramic products, such as lightweight expanded clay aggregate. In Table 1, the main thermophysical properties of the most commercially popular conventional insulation materials are given. Alternative insulation materials can be generated from plants or recycled waste materials, such as coconut husk and bagasse cotton and textile waste, as well as leather, rubber, and plastic waste. Regarding the European insulation material market, mineral wools, namely glass and rock wools, and plastic foams, namely polyisocyanurate, expanded polystyrene, extruded polystyrene, and polyurethane, account for around 58% and 41% of the building thermal insulation market requirement, respectively [28].

**Table 1.** Thermophysical properties of traditional insulation materials (data were retrieved from Refs. [29,30]).

| Insulation Material                 | Density [kg/m <sup>3</sup> ] | Thermal Conductivity [W/(m·K)] | Specific Heat Capacity [J/(kg·K)] |
|-------------------------------------|------------------------------|--------------------------------|-----------------------------------|
| Cellulose                           | 30–80                        | 0.037–0.042                    | 1300–1600                         |
| Wood fibers                         | 50–270                       | 0.038–0.050                    | 1900–2100                         |
| Cork                                | 110–170                      | 0.037–0.050                    | 1500–1700                         |
| Polyisocyanurate (PIR)              | 30–45                        | 0.018–0.028                    | 1400–1500                         |
| Expanded polystyrene (EPS)          | 15–35                        | 0.031–0.038                    | 1250                              |
| Extruded polystyrene (XPS)          | 32–40                        | 0.032–0.037                    | 1450–1700                         |
| Polyurethane (PU)                   | 15–45                        | 0.022–0.040                    | 1300–1450                         |
| Phenolic foam                       | 40–160                       | 0.018–0.024                    | 1300–1400                         |
| Glass wool                          | 15–75                        | 0.031–0.037                    | 900–1000                          |
| Rock wool                           | 40–200                       | 0.033–0.040                    | 800–1000                          |
| Expanded perlite                    | 80–150                       | 0.040–0.052                    | 900–1000                          |
| Expanded vermiculite                | 30–150                       | 0.062–0.100                    | 800–1100                          |
| Lightweight expanded clay aggregate | 290–750                      | 0.08–0.200                     | 900–1000                          |
| Mineralized wood fibers             | 320–600                      | 0.060–0.107                    | 1800–2100                         |

### 2.2. Novel Insulation Materials

The insulation material products that are classified as advanced or superinsulating products include metallic or metalized reflective membranes with gas-filled or evacuated space and aerogels [28]. Except for the gas-filled panels, vacuum-insulating panels (VIPs) and aerogels have been commercially available for almost two decades [31]. However, their total share in the thermal insulation market is restricted to less than 1% [28]. Figure 4 illustrates various forms of commercially available superinsulating materials, including VIPs, monolithic aerogels, silica aerogel granulate, blankets, boards, and paste-like wet mix formulations, for instance, insulating renders.

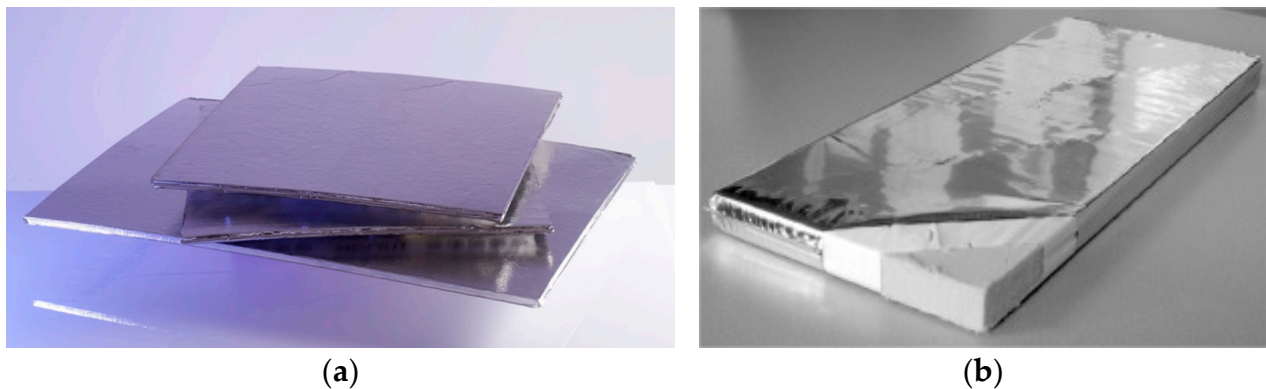


**Figure 4.** (a) Cross-section of a vacuum insulation panel, (b) monolithic aerogel board, (c) silica aerogel granulate, (d) silica aerogel blanket, (e) silica aerogel board, and (f) silica aerogel render (data were retrieved from Ref. [31]).

#### (a) Vacuum-insulating panels

Vacuum-insulating panels (VIPs) are innovative, with a low environmental footprint and high energy efficiency, thermal insulation materials, characterized by an extremely low thermal conductivity value that ranges between  $0.003 \text{ W}/(\text{m}\cdot\text{K})$  and  $0.004 \text{ W}/(\text{m}\cdot\text{K})$  [32]. Due to their exceptionally high thermal resistance capacity, which is almost five times higher in comparison with conventional insulation materials, these panels are constructed to be thin and lightweight [33]. A vacuum-insulating panel is a composite material that consists of a core with a porous structure that allows air evacuation, wrapped by an air and vapor-tight film barrier that air seals the construction and secures its thermal insulating properties [34]. The most commonly used porous core materials concern fiber glass and fumed silica [35], while typical choices also concern polyurethane foam, polystyrene foam, expanded cork, expanded perlite, precipitated silica, and aerogels [33]. Regarding the air and vapor-tight film barrier, aluminum and metalized multilayer foils are used [36]. VIPs' thermal properties degrade over time due to water vapor and other gas penetration through the film barrier, and therefore lead to the deflation of the core [32]. Environmental conditions of high alkalinity, humidity, and temperature can accelerate the degradation of VIPs [32]. In Figure 5, two examples of vacuum-insulating panels are depicted.

The global VIP market was estimated at USD 8.2 billion in the year 2023 and is predicted to amount to USD 11 billion by 2030, growing at a compound annual growth rate of 4.3% over the period 2023–2030 [37]. VIP applications concern the cold chain transport sector and building construction [38]. However, these panels are very fragile and non-receptive to on-site geometry adjustments since the perforation of these panels greatly diminishes their thermal properties [36]. Additionally, the use of joints and supporting components for the installation of the panels induces thermal losses due to the creation of thermal bridges [39]. As a result, only one-fifth of the worldwide VIP applications concern the building sector [40].



**Figure 5.** (a) Different sizes of vacuum-insulating panels and (b) an illustration of the construction material of a VIP (data were retrieved from Refs. [41,42]).

Multiple studies have been conducted testing the application of VIPs on buildings aiming to enhance their thermal performance. For instance, a study by Yuk et al. [43] concerns the energy retrofit of a historic building located in Korea with the application of vacuum insulation panels as external roof insulation. The VIPs are characterized by a thermal conductivity value of  $0.0034 \text{ W/(m}\cdot\text{K)}$ , resulting in the decrease in the roof construction total thermal transmittance by 88.0%. The induced yearly energy savings for winter and summer, concerning the building top floor, were found to be equal to 55.0% and 29.9%, respectively. Moreover, in a study by Zhang et al. [44], the energy saving potential of external wall insulation with VIPs is investigated for the various climatic zones of China. They concluded that the use of VIP panels can be significantly efficient for the colder climates, presenting a decrease in the annual energy consumption which is up to 68.7% and 37.7% higher for the cities of Harbin and Beijing, respectively, in comparison with conventional insulating solutions.

Following these, Uriarte et al. [45] investigated the integration of two opaque external wall energy retrofit solutions, namely, a ventilated façade and a stud-mounted internal system, that incorporate VIPs. The examined buildings concern two large tertiary buildings of  $2100 \text{ m}^2$  area, located in Malmö, Sweden, and Bilbao, Spain, both with heating prevailing needs, and the monitored decrease in the buildings' total energy consumption was found to equal 36.0% and 23.0%, respectively. Finally, Alam et al. [46] simulated the energy renovation of the entire opaque thermal envelope of large office buildings of  $1600 \text{ m}^2$  in the cities of London, Madrid, and Toronto to evaluate the energy savings and payback period of VIPs and other conventional insulation materials. They concluded that the total energy savings were between 6.8% and 10.6%. However, for the VIP solution to be economically viable, a holistic assessment needs to be conducted. Specifically, due to the small thickness of vacuum-insulating panels, there are savings in available living space when using internal insulation techniques, and less construction material is needed to build an equivalent building volume for new constructions. These factors make VIPs an economically viable solution.

### (b) Aerogels

Aerogels are innovative materials characterized by high porosity, low density, and high specific surface area [29,47]. Aerogels are synthesized via the so-called sol–gel process, in which a solution is subjected to a gelation procedure to transform into gel [34]. Through the supercritical drying process, the liquid inside the gel is replaced by air, creating a material of nanoscale porous solid structure [48]. Due to their distinctive and unique structure, aerogels have immense uses in multiple fields of industry, including the building thermal insulation sector, the biomedical sector, the chemistry sector through the formation of catalyst supports and adsorbents, the aeronautics and astronautics sector, and so on [49]. The global aerogel market size was estimated at USD 1.04 billion in 2022 and is predicted to grow at a compound annual growth rate of 16.3% until 2030 [50].

Aerogels can be created from various materials including silica, alumina, chromium, tin oxide, and carbon [34]. Regarding the thermal insulation industry, silica aerogels are used. These materials are characterized by a high porosity that ranges between 85% and 99.8% [48], as well as extremely low thermal conductivity values that in the case of a monolithic material are equal to 0.0131 W/(m·K) up to 0.0136 W/(m·K) [51]. The excellent insulating properties of aerogels derive from their nanoscale pores or cell size of 2 nm up to 50 nm [52]. The air trapped within the pores or cells cannot transmit heat through convection because of the greater viscous drag of air against the cell walls. To further decrease heat transfer, air can be replaced with lower conductivity gases, such as the noble gas argon, or air pressure within the pores' structure can be decreased, as in the case of VIPs [52]. Regarding the building applications of aerogel insulation, two different uses can be detected, either regarding the sole exploitation of silica aerogels' high thermal resistance, as in the case of silica aerogel blankets, with the use of granular aerogel semitransparent insulation materials, or the use of transparent monolithic aerogel, which is extremely brittle.

Multiple studies have investigated the application of aerogels as a thermal insulation material in building constructions. More specifically, Bashir and Leite [53] examined the addition of aerogel silica blankets in a two-story residential building for the tropical climatic conditions of Nigeria. The examined three configurations involved the addition of insulation on the opaque façade either externally or internally, and the internal surface of the attic. They concluded that the latter configuration resulted in the highest energy savings of 15.0% while presenting the best thermal comfort performance and a 6.0% decrease in the recorded indoor air temperature. Furthermore, in a study of Song et al. [54], they developed a sandwich aerogel composite with very low thermal conductivity values of around 0.015 W/(m·K) and 0.017 W/(m·K) and investigated the application of thermal insulation on a building. Specifically, they examined the application of the developed sandwich aerogel insulation and of the two conventional insulation materials of rockwool and expanded polystyrene on a large multi-story residential building for the climatic conditions of different Chinese cities. They concluded that the developed aerogel composite outperformed the conventional insulation solution by up to 31.0% for the heating season and by up to 27.3% for the cooling season.

Moreover, Yin et al. [55] examined the thermal performance of a concrete stadium dome, with 28% of its roof being semitransparent for illumination reasons. A combination of insulation materials was used including outer and inner polytetrafluoroethylene membranes, glass wool blankets, and translucent aerogel, with a transmittance value of 94%. According to numerical and recorded data, the combined insulation configuration results in a 7.7 °C decrease in the average indoor and outdoor temperature difference. Following this, Yue et al. [56] developed an aerogel-based material with high thermal and optical properties. Specifically, they developed a superhydrophobic cellulose aerogel material with self-cleaning capability, characterized by a total thermal conductivity of 0.028 W/(m·K) and a total emittance and reflectance of 91% and 93%, respectively. The application of the developed material was simulated and contradicted the simpler cases of using either an insulation material of the same conductivity or a radiative coating of the same emittance. They concluded that the aerogel composite material can result in 43.4% of extra cooling energy saving for the building sector of China. Lastly, Liu et al. [57] constructed innovative silica-hybridized cellulose acetate aerogel coolers with high thermal insulation and radiative cooling properties. Specifically, the developed material demonstrates a total solar reflectance of 96.0% and a total emittance of 97.0%, as well as a cooling ability of a sub-ambient temperature drop of around 9.15 °C. The application of aerogel coolers on a building rooftop can achieve mean annual energy savings of around 13.90 kW/m<sup>2</sup> for the city of Nanjing in China.

### 2.3. Techno-Economic Comparison of Insulation Materials

Thermal resistance, referred to as R-value, is the most appropriate quantity to assess the thermal performance of a building component [58]. The higher the thermal resistance value



of a building layer, the higher its insulating level. The R-value of a building construction layer is defined as the ratio of its thickness over the thermal conductivity of the material, expressed in  $(\text{m}^2 \cdot \text{K})/\text{W}$ , according to the equation below:

$$R = \frac{t}{\lambda} \quad (1)$$

where  $t$  stands for thickness in m, and  $\lambda$  for thermal conductivity in  $\text{W}/(\text{m} \cdot \text{K})$ . The inverse of thermal resistance is referred to as thermal conductance [58], and in the present study, it is denoted as  $U_{\text{cond}}$ , calculated in  $\text{W}/(\text{m}^2 \cdot \text{K})$  as:

$$U_{\text{cond}} = \frac{\lambda}{t} \quad (2)$$

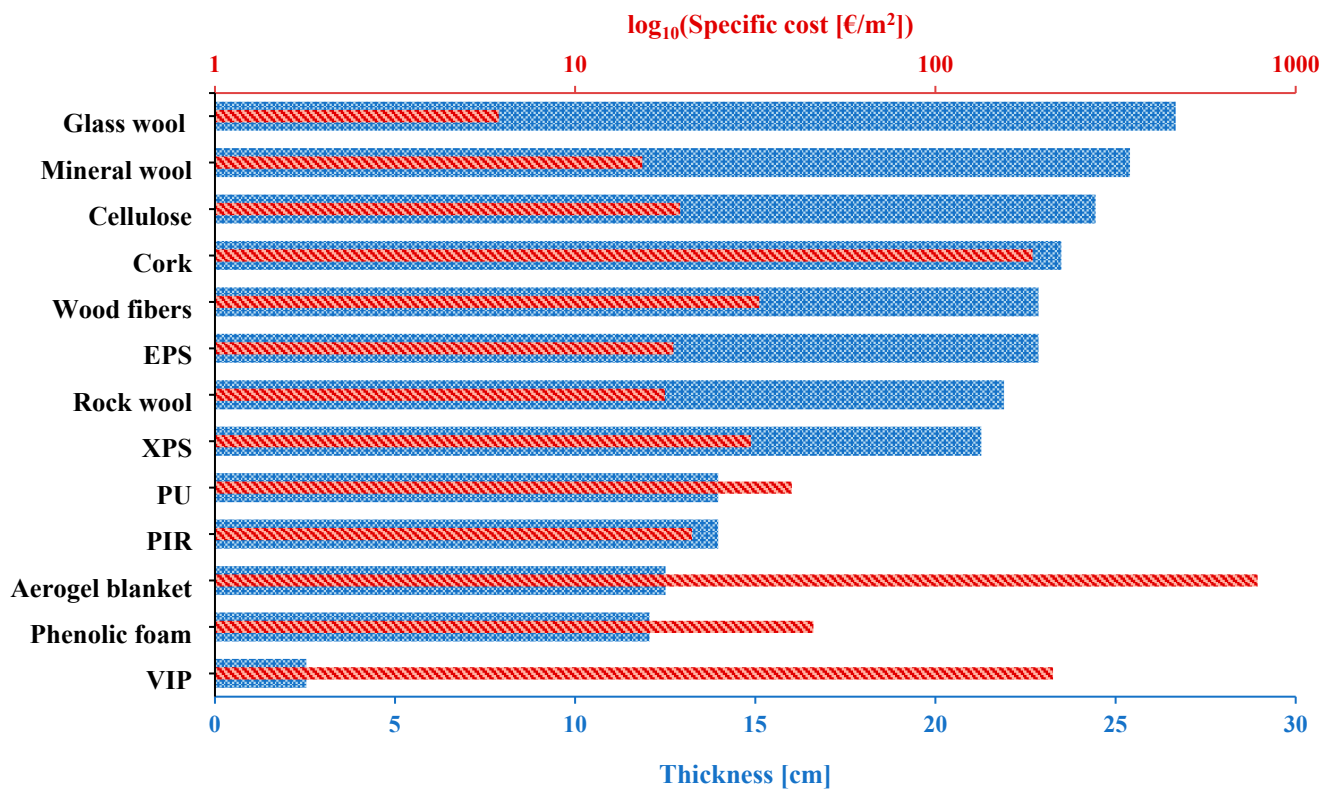
This value does not take into consideration surface resistances or thermal transfer through convection but only through conduction. Table 2 compares multiple insulation materials with respect to the thermal conductivity and cost per area per thickness of the materials. The information regarding the cost is based on today's exchange rate.

**Table 2.** Comparison between insulation materials regarding thermal conductivity and cost.

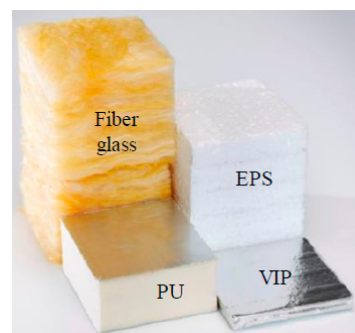
| Insulation Material             | Thermal Conductivity<br>[W/(m·K)] | Cost per Area per<br>Thickness [EUR/m <sup>2</sup> /cm] | Low<br>Flammability | References |
|---------------------------------|-----------------------------------|---------------------------------------------------------|---------------------|------------|
| Glass wool or fiberglass        | 0.040–0.044                       | 0.22–0.24                                               |                     | [59,60]    |
| Rock wool                       | 0.034–0.035                       | 0.74–0.88                                               | ✓                   | [61,62]    |
| Mineral wool                    | 0.037–0.043                       | 0.56–0.65                                               | ✓                   | [63,64]    |
| Cellulose                       | 0.038–0.039                       | 0.78–0.82                                               |                     | [65,66]    |
| Phenolic foam                   | 0.019                             | 3.8                                                     | ✓                   | [67,68]    |
| Expanded polystyrene (EPS)      | 0.036                             | 0.82                                                    |                     | [69,70]    |
| Extruded polystyrene (XPS)      | 0.033–0.034                       | 1.41–1.48                                               |                     | [71,72]    |
| Expanded vermiculite            | 0.62–0.1                          | 2.85                                                    | ✓                   | [73]       |
| Cork                            | 0.037                             | 7.79–8.04                                               |                     | [74,75]    |
| Wood fibers                     | 0.036                             | 1.42                                                    |                     | [76]       |
| Polyisocyanurate (PIR) board    | 0.022                             | 1.52–1.54                                               |                     | [77,78]    |
| Polyurethane (PU) board         | 0.022                             | 2.86                                                    |                     | [79]       |
| Vacuum insulation panel (VIP) * | 0.004                             | 212.1 [€/m <sup>2</sup> ]                               | ✓                   | [80]       |
| Aerogel blanket                 | 0.0197                            | 62.64                                                   | ✓                   | [81]       |

\* Vacuum-insulating panels thickness is equal to 2.54 cm and the cost per area per thickness is defined as [EUR/m<sup>2</sup>].

Next, Figure 6 illustrates the thickness and the specific cost for an area of 1 m<sup>2</sup> per insulation material for an R-value of 6.35  $(\text{m}^2 \cdot \text{K})/\text{W}$  or an equivalent  $U$ -value<sub>cond</sub> of 0.154  $\text{W}/(\text{m}^2 \cdot \text{K})$ . The materials are presented in descending order by thickness, with glass wool and vacuum-insulating panels characterized with the highest and lowest thickness, respectively. In terms of specific cost, glass wool demonstrates the lowest specific cost of 6.13 EUR/cm, while aerogel blankets have the highest equal to 783.6 EUR/cm. Expanded vermiculite is not included in the respective figure due to its relatively much higher values, namely a thickness of 514 cm and a specific cost of 1465.9 EUR/m<sup>2</sup>. If low flammability is taken into consideration, mineral wool and rock wool insulation materials, characterized by a specific cost of 15.37 EUR/cm and 17.75 EUR/cm, respectively, are the most economically viable solutions. Figure 7 depicts the thickness of four insulating materials (fiber glass; expanded polystyrene, EPS; polyurethane, PU; vacuum-insulating panel, VIP) that present the same thermal resistance.



**Figure 6.** Thickness and specific cost of insulation materials for  $U_{\text{cond}}$  of  $0.154 \text{ W}/(\text{m}^2 \cdot \text{K})$ .



**Figure 7.** Comparison of various insulation materials' thicknesses to achieve the same thermal resistance (data were retrieved from Ref. [82]).

#### 2.4. Trends and Challenges of Insulating Materials

The application of insulating materials is an integral action of an energy-efficient building strategy that aims at the minimization of energy consumption for heating and cooling, and the improvement in indoor thermal conditions. The selection of thermal insulating materials is based on the thermal conductivity property of the material, its specific cost, its resistance to fire, and its environmental footprint. Fire-resistant mineral wools, namely glass and rock wools, account for 58% of the European insulation material market, while flammable plastic foams, namely polyisocyanurate, expanded polystyrene, extruded polystyrene, and polyurethane, account for a lower ratio of 41% of the thermal insulation market requirement, respectively [30]. The global market of novel insulating materials, such as aerogels and VIPs, demonstrate high compound annual growth rates, and a slow expansion in building applications. Despite their advanced thermal properties, their high specific cost and installation challenges hinder their widespread application in buildings.

### 3. Innovative Window Technologies

Windows are a building's connection to the outside environment, allowing the passage of daylight illuminance, and direct solar heat gains, and therefore, directly affecting the building energy equilibrium and thermal and visual comfort conditions. The design, orientation, and window-to-wall ratio in a building's geometry are the main parameters that considerably affect the energy consumption and indoor living conditions in a building [83]. However, glazing areas are the most energetically weak aspects in the design of new zero-energy buildings and the deep renovation of the existing building stock [84]. According to the recent literature, a great share, up to 40% or even 60%, of energy losses in buildings on a worldwide scale, is attributed to the poor energy performance of window systems [85]. The predominant type of glass used in the building sector is float glass [86], a flat type of glass characterized by a great level of uniformity on a microscopic level [87]. Float glass is produced through the Pilkington process, in which molten glass is solidified on a bath of molten metal, typically tin [87]. Float glass is characterized by a content of Fe oxides, which is responsible for absorbing solar irradiation at a spectrum range of around  $1\ \mu\text{m}$  [86]. Manufacturing ultra-clear glass is of pivotal essence for the transmittance of visible light into the building's living spaces.

The energy enhancement of windows concerns their thermal and optical properties. The thermal reinforcement of windows can be achieved through the use of multiple glazing panes, triple-glazed or quadruple-glazed windows, with the additional use of noble gases in between the windows, as well as the use of thermal insulators in the windows' frame and spacers [87]. According to the global energy-efficient windows market, double-glazed windows prevail over triple- and quadruple-glazed window systems, accounting for a share of 64.6% in buildings [88]. Regarding the optical properties and the filtering of the incident solar irradiation, three categories of technologies can be discerned, namely, static low emissive coatings, dynamic smart glass chromic technologies, and active kinetic shading devices incorporated in the window's configuration [89]. These window technologies and applications are thoroughly discussed in the next sections. Additionally, a techno-economic comparison of various window systems is conducted by retrieving cost information from the web and local market.

#### 3.1. Multi-Glazed Windows and Low-Emittance Coatings

Due to the fact that heat losses through thermal radiation account for around two-thirds of the total thermal losses of a multiple-glazed window [90], low-emittance coatings' application in the building sector is a widespread energy efficiency technique [91]. According to Granqvist [87], solar energy materials are advanced materials with customized properties regarding the spectral distribution, intensity, and angle of incidence of electromagnetic solar irradiation. Solar energy materials are appropriate to meet the specialized requirements of thermal and electrical conversion applications, as well as energy-efficient passive building technologies, such as windows [92]. The use of appropriate coatings for glazing can enhance their static optical properties and mitigate heat losses through radiation. Thin films characterized by low emittance (low-e) in the spectrum of thermal radiation can result in the restriction of energy losses through windows. For instance, in the case of vertical windows, the use of low-e coatings in double-glazed windows can decrease heat transfer from  $1.5\ \text{W}/(\text{m}^2\cdot\text{K})$  up to  $3.0\ \text{W}/(\text{m}^2\cdot\text{K})$ , while for triple-glazed windows, the corresponding decrease is approximately equal to  $1.0\ \text{W}/(\text{m}^2\cdot\text{K})$  up to  $1.8\ \text{W}/(\text{m}^2\cdot\text{K})$  [87].

Low-e coatings selectively block infrared radiation but are transparent for visible light. Low-e coatings of building applications are ultra-thin metal-based films, mainly of silver, constructed usually through the process of sputter deposition [86]. The sputtered coatings, or soft coatings, are characterized by an emittance value of 10%, while a hard coating, manufactured through the process of pyrolysis, has an emittance of around 15% [93]. On the other hand, a typical uncoated glass is characterized by an emittance value of 84% and a solar transmittance value of only 83.7% [93]. The impurities of mainly salts of iron that exist in soda-lime glass absorb approximately 8.8% of the incident solar irradiation, while

the other 7.5% is attributed to reflection at glass–air interfaces [93]. Conversely, the purest form of glass is referred to as quartz or non-iron glass, and its solar transmittance is around 92% [93]. Low-e coatings have been commercially available for more than 40 years [94] and have captured more than 50% of the global market share [95].

The application of low-e coatings on window configurations is a well-studied energy retrofit action that mainly concerns the combinational use in multiple paned windows. In a multi-objective evaluation study of various retrofit strategies of a single-family residential building in Athens, the replacement of window systems with double- or triple-glazed low-e windows, as a standalone action, is calculated to induce yearly primary energy savings of 2.66% and 5.81% [96]. If combined with the installation of an air-to-air heat pump for heating and cooling coverage, the yearly energy saving rises to 54.40% and 55.82%, respectively. Window replacement is found to belong among the most economic and energy-saving optimum retrofit solutions [96]. Somasundaram et al. [97] investigated the energy-saving potential of an alternative and economically profitable solution of adapting a hard, low-emittance coated glass on the internal surface of glazed surfaces without the creation of an airtight seal, as in the case of typical double-glazed units. Regarding the tropical climatic conditions of Singapore, the investigated configuration was found to result in up to 7.5% annual energy savings for the case of grey-tinted glass, and up to 4.0% for the case of clear glass. Additionally, the illuminance levels were found to be reduced by 75.0%, diminishing the conditions of optical comfort, in contrast to thermal comfort conditions, and the mean radiant temperature of the building which was reduced by 1.3 °C. A similar approach was followed in the energy retrofit of a historic building in Korea, where due to the necessity to preserve the building structure, double-glazed windows with a PVC frame, a 14 mm air layer, and two 4 mm glass panes were added into the internal surface of the existing glazing surfaces [98]. The addition of the double-glazed windows with a total thermal transmittance of 1.94 W/(m<sup>2</sup>·K) and a solar heat gain coefficient of 0.76 resulted in almost 26.0% monthly energy savings.

Additionally, De Masi et al. [84] conducted a thorough study regarding the selection of window parameters with respect to climatic conditions. According to their conclusions, for warm summer, humid, continental and Mediterranean climatic conditions low-e double-glazed windows can result in annual energy savings of 27% in Helsinki in Finland, 57% for Benevento in Italy, and 44% for Split in Croatia. Conversely, for oceanic and continental climatic conditions, like in the case of Oslo, Norway, the energetically optimum solution is the low-e triple-glazed windows, with possible annual energy savings of up to 35%. Finally, Urbikain [57] investigated the combinational use of multiple-glazed air- or gas-filled windows with low-e coatings for the cool climate of Berlin, Germany, and the warm climate of Bilbao, Spain. Regarding the case of Berlin, the option of argon-filled triple-glazed low-e windows outperformed the option of Krypton-filled and air-filled triple-glazed low-e windows. Conversely, in the case of Bilbao, low-emittance double-glazed windows are more efficient than low-emittance triple-glazed windows with regard to annual thermal performance.

Heat transfer through window installations can be restricted through the use of multiple, low-conducting, glass panes. Additionally, the use of transparent insulation between the window panes is an effective technique for eliminating the factors of convection heat transfer through air. More specifically, this can be achieved through (i) the use of solid transparent materials that separate the space between the window panes into numerous cells with dimensions in the scale of 1 cm, or (ii) the use of low-conducting materials, such as noble gases, or finally (iii) the reduction in gas pressure within the window installation [93].

The incorporation of solid transparent insulating materials in a window configuration is another solution to the enhancement of windows' thermal properties. The categories of solid transparent insulating materials that do not perturb vision include flexible polymer foils and inorganic microporous materials, for instance, silica aerogels [93]. Silica aerogels are characterized by low conductivity and high transmittance in the visible region of light and are discerned in two commercially available types for building applications, monolithic

and granular [99]. Monolithic silica aerogels have a higher transmittance value than granular silica aerogels. More specifically, a 1 cm thick monolithic aerogel window presents a total solar transmittance of around 0.9, in contrast to a granular silica aerogel which presents a total solar transmittance of around 0.5 [100]. Additionally, according to experimental studies by Buratti and Moretti [101], regarding energy performance, monolithic aerogel and granular aerogel windows outperform low-e double-glazed windows by 55.0% and 25.0%, respectively.

Regarding the incorporation of solid insulating materials in windows' configuration, various studies have been conducted examining their energy efficiency potential in buildings. For instance, Zhang et al. [102] investigated and compared the thermal performance of ten innovative double- or triple-glazed window configurations that incorporate silica aerogel or PCM, with the thermal performance of a traditional air-filled window configuration, under the severe cold climatic category of China. Through a parametric examination of the optical properties of glass, the thickness parameter silica aerogel, and the PCM melting point, they concluded that the use of the triple-glazing window configuration with a 0.012 m thickness of silica aerogel and a PCM with a melting temperature of 283 K results in a thermal losses reduction of 56.67% compared with a conventional triple-glazing window configuration. Following this, Ramana and Saboor [103] examined the application of a double-glazed window filled with colored hydrogel granules, exploiting its optical filtering property and high reflectance of wavelengths over the near-infrared spectrum, as well as its high thermal capacity. They concluded that for a building model of a total volume of 56 m<sup>3</sup> and the temperature climatic conditions of India, the double-glazed, orange hydrogel granule-filled window system resulted in a 31.7% reduction in the summer solar heat gains.

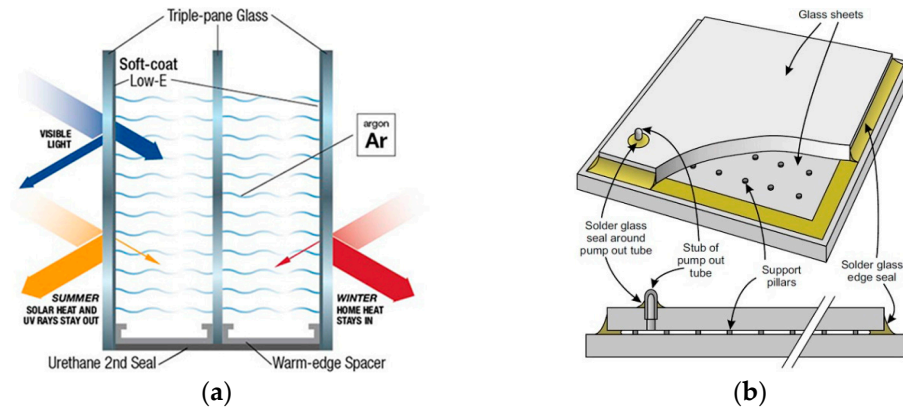
Huang et al. [104] examined a novel double-glazed window configuration that contains a colored transparent aerogel of 16 mm thickness, for the various climatic conditions of China. The window's total thermal transmittance is equal to 0.94 W/(m<sup>2</sup>·K), while the visible and solar transmittance is equal to 44.0% and 78.0%, respectively. According to their energy analysis, the colored transparent aerogel window system demonstrated up to 90.0% energy-saving potential in contrast to single-pane tinted windows in severe cold climatic conditions. Additionally, Mohammad and Ghosh [105] examined the application of double-glazed windows with integrated monolithic or granular aerogel in a two-story residential building in Falmouth in the United Kingdom. Monolithic aerogel outperformed granular aerogel with respect to energy performance and natural illuminance, resulting in a heating energy demand reduction of 15.5% compared to typical double-glazed windows.

Following on from this, the use of noble gases, instead of air, in between the window panes demands the construction of hermetically sealed glazing configurations, which can effectively improve the thermal properties of windows [93]. More specifically, the heat transfer through convection across a typical glass pane distance of 1.2 cm can be reduced by around 10% if air is substituted with argon, while better performances can be recorded with the use of krypton or xenon [87]. If multiple-pane windows are combined with low-emittance coatings, then the energy performance of the windows can be enhanced by up to 20% [87]. A triple-glazed argon-filled window with low-e coatings is illustrated in Figure 8a. However, the introduction of multiple glazing layers, despite the better insulating properties, results in thicker, heavier, and more expensive constructions [106].

An alternative solution is the use of vacuum-insulating windows, as depicted in Figure 8b, which are composed of two glazing layers separated by tiny support pillars, of less than 0.5 mm in length [106]. The construction of vacuum-insulating windows is very challenging and more advanced than the hermetically sealed configurations of gas-filled windows [107]. The gas pressure should be less than 10<sup>−6</sup> of atmospheric pressure, which requires the presence of hermetic edge seals. Additionally, the support pillars used to maintain the separation between the glazing layers must withstand the mechanical pressure exercised by the atmospheric pressure, and be resistant to heat transfer between the glazing layers. Finally, the frame of the vacuum-insulating window must be receptive to movements due to thermal expansions of the glazing layers [107]. Despite the technical



challenges of vacuum-insulating windows, they manage to eliminate heat transfer through conduction and convection. Indicatively, vacuum-insulating windows can be characterized by a total thermal transmittance value of less than  $1 \text{ W}/(\text{m}^2 \cdot \text{K})$  [108]. The combinational use of vacuum-insulating windows with low-e coatings can also eradicate heat transfer through radiation [109].



**Figure 8.** (a) Triple-glazed argon-filled window with low emittance coatings, and (b) schematic diagram of a vacuum-insulating window (data were retrieved from Ref. [110]).

The technical challenges posed in the construction of vacuum glazing due to the edge-seal effects, in combination with the thermal losses at the edges of windows that considerably increase the configuration's total thermal transmittance, have set the application of vacuum glazing as an energetically and economically non-optimum solution [111]. Therefore, vacuum glazing and its combinational use with other technological solutions have drawn the attention of the scientific community. For instance, photovoltaic glass is combined with vacuum glazing to enhance the thermal insulation level of buildings, restricting solar heat gains, facilitating adequate levels of daylight illuminance, and contributing to building energy self-sufficiency through the production of electricity [112]. In a study by Uddin et al. [112], a system of semitransparent photovoltaic glass in combination with vacuum glazing is investigated for five climatic categories in China. Regarding the energy-saving potential, the investigated system is proved to be more efficient for cooling-dominated cities, resulting in an up to 76.3% decrease in annual energy consumption, and an up to 59.4% decrease in heating-dominated cities.

### 3.2. Chromogenic Materials

Chromogenic materials are a smart window technology, characterized by a reversible adaptation of windows' optical properties to external stimuli [113]. Two emerging technologies of chromogenic materials concern thermochromic and electrochromic materials and can be used to enhance energy efficiency in buildings by controlling incoming solar irradiation, visible light, and thermal radiation [114]. In a few words, thermochromic thin films, using vanadium dioxide, reduce solar energy transmission at high temperatures compared to low temperatures, while electrochromic devices, typically based on tungsten oxide and nickel oxide, alter their solar and visible light transmittance when voltage is applied [114].

#### 3.2.1. Thermochromic Films

Vanadium dioxide is the most promising of the known thermochromic materials and can be used as a basis for the construction of thermochromic glazing [114]. Vanadium dioxide thin films present abrupt changes in their resistance at a specific critical temperature when a change in the material structure is realized. Specifically, when the temperature of the material is below the critical temperature, which for vanadium dioxide is at  $68^\circ\text{C}$ , the monolithic structure of the materials demonstrates semiconducting properties and a high value of spectral transmittance at high infrared wavelengths. Conversely, when the temper-

ature of the material is over the critical temperature, the rutile structure demonstrates high conductivity and infrared reflectance. The critical temperature of 68 °C can be decreased to a more suitable one for building applications if a small percentage of vanadium atoms is replaced by tungsten [115]. Furthermore, additives such as magnesium, zinc, and fluorine can be used to enhance the transmittance of visible light of the thermochromic films [116]. Finally, the use of thermodynamically stable materials, such as vanadium pentoxide, or overcoating such as with aluminum oxide or aluminum nitride, can prolong the life cycle of thermochromic films even at higher temperature and humidity levels [117]. The construction of thermochromic windows can be achieved with the methodology used for the application of thin films upon insulated glass units with static properties [118].

Teixeira et al. [119] investigated the visual, thermal, and energy advantages of thermochromic double-glazed windows in comparison with conventional double-glazed windows, for the various European climatic conditions, in an office room according to simulation and measurement data. They concluded that thermochromic glazing can result in superior indoor conditions, of almost 90.0% daylight illuminance utilization, 50.0% thermal comfort conditions, and a 50.0% decrease in energy consumption in contrast to clear glass. Another study concerning the application of thermochromic glazing in office buildings with an increased glazing surface ratio was conducted for the climatic conditions of the cities of Cairo, Barcelona, and London [120]. The parametric study regarding the thickness of the glazing revealed an energy consumption decrease of up to 38.4%, 46.4%, and 33.3% for the desert-type climate of Cairo, Mediterranean-type of Barcelona, and temperate-type of London, respectively. Finally, Khaled et al. [121] performed a comprehensive thermal evaluation of commercially available thermochromic glazing by taking into consideration spectral selectivity, the phenomenon of hysteresis in the color and transmittance switching process. In contrast to clear and low-e glazing, thermochromic glazing can result in yearly specific energy reductions of up to 6.3 kWh/m<sup>2</sup> and 12.0 kWh/m<sup>2</sup> for the climatic conditions of Toronto and Abu Dhabi, respectively [113].

### 3.2.2. Electrochromic Films

Electrochromic windows reversibly change their optical property of transmittance in response to voltage. Electrochromic multilayer coatings are composed of five layers positioned between two glass substrates in a laminate configuration [122]. The outermost layers are transparent electrical conductors that allow insertions and extractions of electrical charge from an external stimulus that controls the reversible modulation of the film transmittance. The central layer is an ion-conducting solid material, which can be either an inorganic or polymeric layer [113]. The central electrolyte layer joins two electrochromic films, one of which is coloring under charge insertion, while the other is coloring under charge extraction [122], in other words, the thin electrochromic oxide films function as an anode and cathode [114]. When a voltage of less than 2 V is applied to the transparent electrical conductors, ions are transferred from the ion storage through the ion conductor and into the electrochromic layer [113]. Except for the aforementioned laminated electrochromic device, a monolithic electrochromic device can be created through the sequential deposition of the five different layers onto a glass pane [114]. However, the latter methodology presents mechanical and chemical disadvantages when applied to insulated glass panes [114]. On the other hand, the laminated construction can be applied to any flat or curved-shaped glazing, while it is receptive to adjustments to any size [122].

The sector of electrochromic windows has experienced gradual progress and important technological innovations have made, apart from being energy efficient, an economically viable solution [114]. The most advanced materials from the construction of electrochromic films include tungsten and nickel oxides, while additives like iron or titanium can increase the durability of the films [123]. In a study by Detsi et al. [124], the combinational use of thermochromic and electrochromic coatings on triple-glazed systems is investigated in the case of a multi-story office building under the climatic conditions of Athens and Stockholm. The concept of the combinational use of chromogenic coatings is concluded to result in

18.5% and 8.1% annual primary energy savings, for the cities of Athens and Stockholm, respectively. Fathi and Kavooosi [125] calculated that the combinational use of smart electrochromic glazing, building-integrated photovoltaic systems, and energy management systems can result in an up to 35.0% decrease in a high-rise building's yearly energy consumption. Finally, Hoon Lee et al. [126] studied and found that the optimal control parameter to modulate the adaptive behavior of electrochromic windows concerning building energy efficiency is the ambient temperature level. In their study, they examined a three-story commercial building for six different climatic categories of the U.S. and concluded that the use of an electrochromic glazing system can result in a 16.5% decrease in the capacity of a building cooling system and an 11.8% decrease in the daily maximum cooling load.

### 3.3. Techno-Economic Comparison of Window Installations

In this section, techno-economic information regarding typical and innovative window systems is given. Specifically, Table 3 summarizes various window systems of aluminum, unplasticized polyvinyl chloride (uPVC), or polyvinyl chloride (PVC) single sash (SS) or double sash (DS). Information about specific costs is derived either from websites or from the local market based on today's exchange rate. According to the cost information presented in Table 3, regarding the construction materials, uPVC or PVC window installations are discerned as a more economical solution compared to aluminum window installations of comparable insulative properties. For instance, regarding the case of a double-glazed window system, the specific cost of aluminum windows with a total U-value of  $1.30 \text{ W}/(\text{m}^2 \cdot \text{K})$  is 760 EUR/m<sup>2</sup>, while the specific cost of uPVC argon-filled windows with a total U-value of  $1.40 \text{ W}/(\text{m}^2 \cdot \text{K})$  is 283 EUR/m<sup>2</sup> (269% lower cost). Additionally, regarding triple-glazed windows, the specific cost of low-e aluminum windows with a total U-value of  $1.10 \text{ W}/(\text{m}^2 \cdot \text{K})$  is 885 EUR/m<sup>2</sup>, while the specific cost of PVC low-e argon-filled windows with a total U-value of  $0.96 \text{ W}/(\text{m}^2 \cdot \text{K})$  is 660 EUR/m<sup>2</sup> (134% lower cost). Moreover, the highly insulative quadruple-glazed windows system is a very heavy and expensive solution. For instance, in comparison to an aluminum low-e triple-glazed window with a total U-value of  $1.10 \text{ W}/(\text{m}^2 \cdot \text{K})$ , an aluminum quadruple-glazed window with a total U-value of  $0.80 \text{ W}/(\text{m}^2 \cdot \text{K})$ , lower by 27.3%, is characterized by a 172.3% higher specific cost. Lastly, restricted cost information was available for the chromogenic window systems, validating that electrochromic and thermochromic windows are still expensive and not widespread options for building applications.

**Table 3.** Comparison of window technologies in terms of total thermal transmittance and cost.

| Description                                                   | Total U-Value [ $\text{W}/(\text{m}^2 \cdot \text{K})$ ] | Specific Cost [EUR/m <sup>2</sup> ] | Reference |
|---------------------------------------------------------------|----------------------------------------------------------|-------------------------------------|-----------|
| Aluminum SS low-e double-glazed side-hinged window *          | 2.10                                                     | 420                                 | [127]     |
| Aluminum DS low-e double-glazed side-hinged window *          | 2.10                                                     | 470                                 | [127]     |
| Aluminum DS double-glazed side-hinged window                  | 1.30                                                     | 760                                 | [128]     |
| uPVC DS double-glazed argon-filled side-hinged window         | 1.40                                                     | 283                                 | [129]     |
| uPVC DS low-e double-glazed argon-filled side-hinged window * | 1.10                                                     | 550                                 | [130]     |
| Aluminum DS low-e triple-glazed side-hinged window            | 1.10                                                     | 885                                 | [128]     |
| Aluminum SS triple-glazed argon-filled side-hinged window *   | 0.6                                                      | 610                                 | [130]     |
| PVC DS low-e triple-glazed argon-filled window                | 0.96                                                     | 660                                 | [129]     |
| Aluminum DS quadruple-glazed side-hinged window               | 0.80                                                     | 1525                                | [131]     |
| Double-glazed electrochromic window                           | 1.30                                                     | 1100                                | [132,133] |
| Thermochromic window                                          | 1.30                                                     | 860                                 | [134,135] |

\* Prices are derived from the local market.

### 3.4. Trends and Challenges of Window Systems

The trends in window systems technology are geared toward enhancing energy efficiency through the incorporation of multiple glazing panes, the application of low emittance coatings, and the use of noble gases. Multi-paned and gas-filled window systems are airtight and heavy configurations, and their thermal behavior is directly linked to their proper and airtight installation on the building envelope. Their specific cost ranges according

to the construction material, with uPVC being the most economically and thermally efficient option. Triple- and quadruple-glazed windows are gaining popularity; however, double-glazed windows remain the most prevalent option. Triple- and quadruple-glazed windows result in energy saving and noise reduction and abide by national and international strict energy regulations. Despite a higher initial cost, the long-term energy savings and enhanced indoor thermal comfort encourage their widespread application, especially in regions with severe weather conditions. Moreover, emerging novel window systems concern chromogenic and vacuum glazing. Vacuum glazing presents technical challenges in its construction due to the edge-seal effects, and a high risk of thermal bridge formation. Therefore, vacuum glazing is still an energetically and economically non-optimal solution. The challenges linked to chromogenic window systems concern their low lifespan and the difficulty in tuning their dynamic behavior. The modulation of the response time of chromogenic glazing is a great challenge that affects both their thermal behavior and their high initial cost.

#### 4. High Thermal Mass Technologies

Construction material's thermal mass determines a building's storage heat capacity, and, subsequently, its indoor thermal conditions, thermal loads, and thermal comfort. High thermal mass technologies are characterized by high sensible or latent heat storage capacities that combat high temperature changes and reduce the rate of change. Thermal mass is particularly important during periods of transient heating and cooling, which is the predominant thermal mode for most buildings worldwide, as very few operate under continuous, steady-state conditions [136]. In this section, the technological solutions of high thermal mass and their innovative applications for building use are presented. Specifically, phase change materials (PCMs) and the architectural design of the Trombe wall are properly analyzed.

##### 4.1. Phase Change Materials

According to the mechanism of energy storage, materials are categorized as (i) materials that store sensible heat over their entire temperature range of use, and (ii) materials that, also, store a large amount of latent heat in a narrow temperature band [137]. Phase change materials are discerned by their phase change temperature range and their composition [138]. Organic PCMs are mostly selected for low- and medium-temperature applications and include paraffins [139], alcohols, fatty acids, and advanced aliphatic hydrocarbons [140]. Organic PCMs are non-corrosive with most materials, chemically and thermally stable [141], less expensive than other available PCMs, and can be produced in various forms, including powder, granules, and plates. Figure 9 illustrates the cases of non-packaged and microencapsulated paraffin. In comparison to inorganic PCMs, organic PCMs are characterized by a lower latent liquefaction heat, thermal conductivity, and density, while they are flammable at relatively low temperatures [142]. Conversely, inorganic PCMs are used in medium- and high-temperature applications and include crystalline hydrated salts [143], molten salts [144], metals, and their alloys [145]. Figure 10 illustrates the cases of non-packaged and microencapsulated hydrated salts. Inorganic PCMs are corrosive and hygroscopic materials that require reinforced packaging [138]. Additionally, their thermal properties are depleted over several charge cycles, restricting the material's lifespan. The main cause of degeneration in PCMs is phase separation and anhydrous salt formation, an irreversible process that occurs because hydrates melt according to the lowest component of the mixture [146]. Another serious problem that concerns all inorganic and a part of organic PCMs is supercooling, which is associated with the material's inefficiency to utilize stored heat [146]. The packaging of PCMs isolates the material from the environment and facilitates the transmission of stored heat [147]. Encapsulating PCMs in spheres under 1 mm in diameter increases heat exchange and controls volume changes during phase transitions [138]. The protective shell is usually plastic or synthetic resin [138].





**Figure 9.** (a) Non-packaged paraffin, (b) microencapsulated PCM with grain composition (data were retrieved from Refs. [148,149]).



**Figure 10.** (a) Packaged and (b) encapsulated salt hydrate PCM (data were retrieved from Ref. [150]).

Phase change materials are widely exploited in fields that require temperature control and heat absorptance or heat abduction at a constant temperature, for instance, in lithium-ion batteries, satellite equipment, electronic components, construction, and textile applications [138]. Regarding building applications, the high energy storage density of phase change materials makes them materials of high thermal mass, the incorporation of which in the building thermal envelope is indicated as an energy efficiency tactic [151]. The integration of phase change materials into the building's envelope as a passive energy efficiency system can improve a building's thermal inertia [152], acting as a heat transfer regulator between the indoor building space and the ambient temperature [153].

The conventional method of applying PCMs in buildings is via adding a layer of material. The most popular PCM used in building applications is paraffin wax, which is inexpensive, but flammable, with a high thermal conductivity material that derives from petroleum, a non-renewable source. An alternative method is using a component of composite construction materials, for instance, cement [154] or ceramic-based components [154], mortar, and concrete [155]. In Figure 11, the cases of clay brick with incorporated PCM and a sample of PCM foamed cement are illustrated. Multiple methods have been tested regarding the incorporation of phase change materials into building materials, including micro-encapsulation, shape stabilization, macro-encapsulation, and porous inclusion. These methods ensure the retention of the material even after its complete condensation, as well as the non-intake of moisture from the ambient air.





**Figure 11.** (a) Clay brick with macro-encapsulated PCM, and (b) PCM foamed concrete sample (data were retrieved from Refs. [156,157]).

Various innovative PCM composite building elements have been constructed, tested, and compared with typical building wall and roof elements. For instance, Abass and Muthulingam [158] experimentally investigated the thermal energy storage performance of a biaxial voided roof slab integrated with PCM under the ambient conditions of India, comparing it with a standard reinforced concrete slab without PCM. Their results indicate that the PCM-integrated biaxial voided roof slab can reduce interior roof temperatures by up to 7.2 K during sunny hours, and decrease the thermal cooling load by up to 54%. Additionally, an innovative concept is the incorporation of PCM into foamed cement or foamed concrete. Foamed cement is a new type of lightweight thermal insulation material, characterized by high porosity, low thermal conductivity, and dry density, with hygroscopic characteristics [159]. Foamed cement is a better insulator but a worse heat capacitor than pure cement. The addition of phase change material into foamed cement results in the creation of a product that combines the advantages of foamed concrete, namely low thermal conductivity, and phase change materials, namely energy storage density. In a study by Meng et al. [157], they experimentally tested three rooftop configurations in two small volume test rooms, in the city of Suzhou in China. The examined rooftops concern the PCM foamed concrete rooftop, the foamed concrete rooftop, and a typical cement roof, while the addition of high-reflectance coating to the aforementioned configurations is also examined. According to their results, the PCM foamed concrete rooftop is the most efficient in maintaining a low indoor air temperature and heat gain. More specifically, in comparison to the typical concrete rooftop, the PCM foamed concrete resulted in a 2.0 K and up to 2.9 K lower mean indoor air temperature, and reduced heat gains by 48.5% and 59.0% for the cases of no and an additional high-reflectance coating, respectively. Interestingly, phase change materials can also be incorporated in asphalt products, as a means of mitigating the heat island effect [160]. The concept of PCM-doped pavement needs further research; however, according to the experiment of Mizwar et al. [161], the heating rate of the composite material and the duration of the peak temperature decreased, and the maximum temperature decrease was equal to 12.0 K in comparison to the conventional asphalt. Another interesting application of PCMs is the technique of PCM-embedded pipes in buildings and underground structures. More specifically, the PCM-integrated and thermally active building envelope stably regulates the thermal and humidity indoor environment of buildings and underground spaces by using heat exchange pipes embedded in structures that utilize low-grade energy and the PCM heat-storage properties [162].

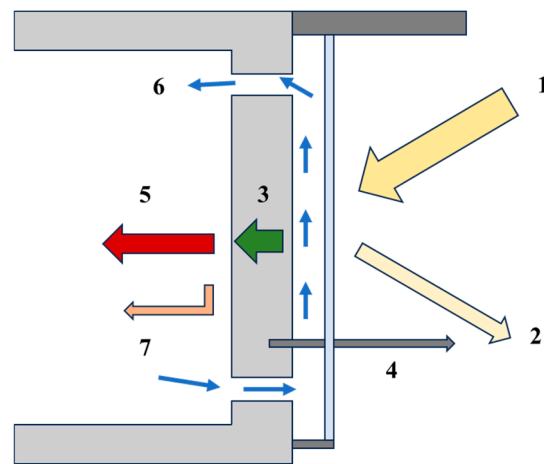
The integration of PCMs in the building's thermal envelope has been thoroughly studied concerning the induced thermal advantages. The external incorporation of PCM on two-floor residential buildings was examined for the climatic conditions of Athens and was found to induce greater heating (1.1–4.2%) than the cooling (0.2–0.7%) energy savings [163]. The two-retrofit strategy of PCM and thermal insulation on the building rooftop demonstrates high synergy for the winter period, resulting in total heating energy savings of 11.7–30.1%. More specifically, according to Beemkumar et al. [164] and

Pasupathy et al. [165], the incorporation of PCM on a building's rooftop modulates the fluctuation of the room's temperature, restricting the mean peak indoor temperature by 1 up to 2 K, therefore enhancing the thermal comfort conditions. The selection of PCM parameters, namely, thickness, latent heat, location on the building envelope, and phase change transition temperature range should be properly adapted to a location's climatic typology to maximize energy savings [166]. For instance, in a study by Nurlybekova et al. [167], where the projected future climatic subtropical conditions of China, Vietnam, and India are considered, a parametric analysis of the PCM phase change temperature, integrated in a two-story office building, is conducted. According to their calculations, the application of the optimum PCMs resulted in a maximum of a 37.0% enhancement in the building's energy performance. Following this, Dardouri et al. [168] calculated that for locations with a Mediterranean climate, the modulation of the PCM layer on the roof has an immediate effect on the building's thermal loads, leading to a yearly energy reduction that ranges between 8.0% and 31.5%. Furthermore, in a study by Panayiotou et al. [169], the implementation of PCM in a Mediterranean dwelling, and its combination with the envelope's thermal insulation, is calculated to provide energy savings of 66.2%, a reasonable payback period of 7 years, as well as an enhanced performance during the summer period with a temperature decrease of 3 up to 5 K [169]. Similarly, Tsoka et al. [170] found that the integration of PCM in a typical Greek residency demonstrates better energy performance and thermal comfort conditions for warmer climatic zones. They also concluded that the proper adjustment of PCM melting temperature for each location is substantial for maximum energy improvement.

#### 4.2. Trombe Wall

The Trombe wall is one of the most widespread and commonly adopted solar passive techniques, useful to enhance a building's thermal performance [171]. As a passive solar system, the architectural concept of the Trombe wall exploits solar thermal irradiation and the building's thermal mass, to store thermal energy, and to release it with hysteresis through radiation, convection, and conduction to the building [172]. It is reported that the adoption of the Trombe wall system can result in a total decrease of up to 30.0% in the building's heating and cooling energy demand. Additionally, the Trombe wall system is characterized by a low initial and operational investment cost, while its lifespan can be considered equal to the lifespan of the building in which it is integrated [83]. According to Figure 12, the typical Trombe wall consists of a solid, south-facing wall of high heat capacity made of concrete, with a thickness value of 30 cm up to 40 cm, painted in a dark color of high absorptance, usually black, on its outer surface. At a close distance of around 3 cm, there is a glass surface and at the top and bottom of the wall, there are vents that facilitate air movement, while there is also a vent in the upper part of the glazing.

The operation of the Trombe wall system is based on the phenomenon of natural ventilation and is carried out by the circulation of air in the space between the glazing and the wall, and through the vents because of the temperature difference. More specifically, during a winter day, the solar irradiation intercepted by the glazing heats the air located between the glass and the wall. The warm air moves upwards and through the vent to enter the interior building space. Simultaneously, the cooler air from the interior enters from the bottom vent and fills the space created. During the night, the operation is reversed, the two vents remain closed, and space heating is achieved by the radiation of the stored heat in the wall. During the summer operation, the top wall vent remains closed, while the vent at the top of the glazing is open, and warm air is rejected to the ambient conditions. Additionally, with an appropriate static shading element, direct solar irradiation can be minimized.



**Figure 12.** Schematic of Trombe wall. 1: Incident solar irradiation. 2: Reflected solar irradiation. 3: Thermal energy stored in the wall. 4: Thermal energy lost by the wall. 5: Thermal energy radiated to the interior with hysteresis. 6: Thermal energy transferred through air movement. 7: Thermal energy transferred from the wall through natural convection.

The typical configuration of the Trombe wall system, however, is reported to result in inconsistent heating during the winter period, and overheating during the summer period [173]; therefore, numerous novel technologies are incorporated in the systems to enhance its performance. The operation of the performance of the Trombe wall systems is determined by several pivotal parameters that regard the thickness and thermal properties of the wall mass, the glazing thermal and optical properties, the area of the Trombe wall, the shading system, and the inter-space or channel depth [174]. The integration of the Trombe wall systems with phase change materials is a technique reported to enhance the efficiency of the system [175]. More specifically, the addition of PCM increases the specific storage and release capacity of the wall within a narrow temperature range due to the phase transition process, which results in better regulation of the indoor air temperature and thermal comfort conditions [175]. For instance, Bellos et al. [176] investigated an innovative Trombe wall configuration with additional windows in the massive wall that exploits a share of direct solar irradiation for space heating and satisfies sufficient daylight illuminance. According to their study, the proposed Trombe wall configuration results in a 0.5 K main air temperature increase during the heating season. On the other hand, aiming to reduce thermal discomfort during summer, Liu et al. [177] examined a PCM–Trombe wall configuration with radiative cooling properties and calculated a 55.2% reduction in indoor overheating during summer in contrast to the simple Trombe wall.

Following this, a dynamic double-skin Trombe wall was proposed, which is an innovative solution that consists of the Trombe wall and a PCM layer that can exchange positions through a rotational move about the system's vertical axis [178]. According to the experimental testing of Zhou and Razaqpur [85], the innovative double-skin Trombe wall system can preserve a constant temperature level in the indoor space by 19.3% greater efficiency and achieve around 29.0% less heat loss through glazing and the solidification process of the PCM, in comparison to the typical Trombe wall system. The Trombe wall system with a PCM layer and an additional insulation layer on each side of the wall was investigated by the same authors through a CFD analysis simulation and compared with a static Trombe wall system with and without PCM [179]. The simulation results indicate that the novel dynamic Trombe wall system with enhanced thermal conductivity and heat capacity is 25.3% and 17.5% more energy efficient, and 79.0% and 35.0% more thermally efficient than the static Trombe wall, with and without a PCM layer, respectively.

The glazing system determines the amount of incident solar irradiation that is reflected, absorbed, and transmitted, as well as the heat transfer between the inter-space and the ambient environment. The incorporation of multi-glazed windows into the Trombe wall

system can reduce energy demand for heating and restrict heat losses during the winter nights, and simultaneously, they reduce solar heat gains and the possibility of overheating during the summer period. Advanced insulation glazing systems, for instance, vacuum-insulating glazing, gas-filled windows, silica aerogel-integrated windows, or low emittance coatings, can be utilized in combination with the Trombe wall system and result in higher energy performance [81]. More specifically, Xiao et al. [180] investigated a ventilated low-e glazing Trombe wall system in a small test room for the climatic conditions of China. They concluded that the examined system can outperform the conventional Trombe wall system by 11.1% regarding the heating energy demand, for the case of an air-conditioned building. In general, the impact of the glazing's thermal and structural properties can properly regulate solar heat inputs and mainly affect the summer period building operation, which can be enhanced by up to 12.2% [181]. Lastly, another examined solution is the integration of photovoltaic–thermal systems and the double exploitation of solar irradiation for heating spacing and power generation [172]. The main advantage of the respective composite systems is that they can adapt to a building's seasonal thermal demands, and in winter, warm air can be inserted into the living space, while during summer, warm water can be produced instead [182].

#### *4.3. Trends and Challenges of High Thermal Mass Technologies*

An innovative solution of incorporating high thermal mass technologies into buildings concerns the creation of PCM composite materials, such as PCM clay bricks and PCM foamed concrete. PCM composite materials are mainly experimentally tested materials that demand highly durable encapsulating techniques to prevent material leakage and thermal degradation of the material properties. Another solution concerns the design of the Trombe wall and its thermal enhancement through the incorporation of energy efficient elements, such as PCMs, thermal insulation, low-e and highly resistive glazing, and a ventilation system. However, the thermal upgrade of the Trombe wall transforms a simple passive solution into a multi-parameter, more expensive, and complicated system.

### **5. Optically Advanced Coatings with Cooling Potential**

The optical properties of the coatings used in the exterior of buildings is a critical parameter for buildings' energy equilibrium, determining the fraction of absorbed and reflected solar irradiation intercepted by opaque building elements. In this section, a thorough discussion regarding cool-colored materials with static or dynamic optical properties and their application in the building sector is conducted.

#### *5.1. Cool Materials with Static Optical Properties*

The application of optically advanced coatings on buildings and urban constructions, characterized by low and very low surface temperature levels, is a well-studied technological field, with a substantial effect on the mitigation of the heat island phenomenon [183]. Regarding building applications, cool roofs are the most examined mitigation solution on buildings, reported to result in a reduction in the energy demand for space cooling, and in the surface temperature and ambient temperature, due to their high solar reflectance, also referred to as albedo [183]. The most popular and commercially widespread cool-colored coatings are the acrylic-based paints for roof applications, demonstrating high solar reflectivity and a small rate of optical degradation over time. However, except for the reflectance in the visible range, technological advancements in the field of cooling materials have addressed optical properties in the entire electromagnetic spectrum, as well as studying the angle of reflection as a parameter of the passive cooling potential as in the case of the retro-reflectivity effect [184], and also the phenomena of fluorescence [185] and thermochromism [186]. The main principles of the design of cooling materials concern (i) low solar absorptance in the visible or infrared spectrum, and (ii) high emissivity in the infrared spectrum, as a means to enhance thermal dissipation through radiation. However, as defined by Planck's law, high emissivity in the entire infrared spectrum equals

high absorbance in the respective wavelengths, resulting in the absorbance of incoming atmospheric thermal radiation [183]. The respective optical disadvantage is addressed through photonic or plasmonic coolers, which are characterized by high emittance in the wavelength spectrum band of 8 and 13  $\mu\text{m}$ , referred to as the atmospheric window. Table 4 summarizes the optical characteristics of multiple technological solutions appropriate for mitigating the urban heat island phenomenon.

**Table 4.** Optical properties of various technological solutions with cooling potential (data were retrieved from Ref. [183]).

| Type of Material                       | Optical Properties                    |                                        |                          |                                      |                           |
|----------------------------------------|---------------------------------------|----------------------------------------|--------------------------|--------------------------------------|---------------------------|
|                                        | High Reflectance in the Visible Range | High Reflectance in the Infrared Range | High Broadband Emittance | High Emittance in 8–13 $\mu\text{m}$ | High Fluorescent Emission |
| Light color reflective                 | ✓                                     |                                        | ✓                        |                                      |                           |
| Colored infrared reflective            |                                       | ✓                                      | ✓                        |                                      |                           |
| Reflective with PCM                    | ✓                                     | ✓                                      | ✓                        |                                      |                           |
| Thermochromic                          | ✓                                     | ✓                                      | ✓                        |                                      |                           |
| Fluorescent                            | ✓                                     | ✓                                      | ✓                        |                                      | ✓                         |
| Photonic and daytime radiative cooling | ✓                                     | ✓                                      |                          | ✓                                    |                           |

A categorization method of cool materials is proposed by Morales-Inzunza et al. [187], according to whom cool materials are discerned as, (i) light-colored materials, (ii) cool-colored materials, (iii) daytime radiative cooling materials, (iv) fluorescent materials, (v) retro-reflective materials, and (vi) thermochromic materials. Light-colored materials include natural and manufactured materials and are characterized by high reflectance in the visible spectrum, which results in lower surface temperature levels in contrast to darker materials [187]. A natural light-colored material is marble, with a total reflectance of 84.0%, and a high specular reflection and disadvantage of glare [183]. Manufactured light-colored materials include ceramic tiles, rooftop membranes, diffuse white paint, and concrete composites [187], as well as mortar mixtures with additive white marble [188].

Cool infrared materials are characterized by high reflectance in the near-infrared band, a component that accounts for almost half of the solar electromagnetic spectrum and for the greatest share of heat absorbed in cities [189]. Compounds that demonstrate high reflectance in the near-infrared spectrum, but also cause severe glare issues, include titanium dioxide, characterized by a near-infrared reflectance of up to 80.0%, and calcium carbonate, characterized by a near-infrared reflectance of up to 70.0% [190]. On the other hand, cool-colored materials can be found as membranes, coatings, or concrete mixtures, and are characterized by high infrared, especially near-infrared, reflectance but not high reflectance in the visible range. Cool-colored materials constitute the second generation of cool materials, according to Becherini et al. [191]. The scientific interest is orientated toward the development of cool-colored materials with colorful tones [192]. Exposure to outdoor environmental conditions induces problems of corrosion, optical degradation, and dirt, leading to the deterioration of cool-colored materials' emittance and reflectance properties [193]. In this direction, the development of cool-colored materials with hydrophobic behavior and self-cleaning properties can decrease the impact of soil and dust deposition and maintain the optical and thermal properties of the materials [194].

Following retro-reflective materials is an alternative option for combating the urban heat island effect due to their ability to reflect incoming irradiation in the angle of incidence [195]. The property of retro-reflectance eliminates the phenomenon of the multi-reflectance of irradiation between the buildings and contributes to reflecting solar irradiation outside the city. According to the existing literature, the most researched



retro-reflective materials are constructed with glass spheres, barium spheres, and prism structures [187].

Daytime radiative materials are characterized by high broadband solar reflectance and high emissivity with the infrared band of 8 up to 13  $\mu\text{m}$ , or the atmospheric window [187]. Daytime radiative materials can decrease their temperature below the ambient level, achieving negative equilibrium, due to the excess of emitted radiation over that absorbed [183]. According to the latest technological advancements in radiative cooling, effective radiative coolers are composed of metamaterial, photonic, and plasmonic structures [183]. The four distinguished categories of photonic radiative coolers include photonic surfaces with a multilayer planar structure, photonic metamaterials, paints designed for daytime radiative cooling, and polymers [196]. The performance of daytime radiative materials is affected by atmospheric conditions, for instance, humidity and pollutants, reducing the transmissivity of radiation at the atmospheric window [197].

Finally, fluorescent materials emit light in the ultraviolet and visible range when absorbing electromagnetic irradiation. The emitted irradiation is characterized by a comparatively longer wavelength to the light absorbed [183]. The difference between the absorption peak wavelength and the fluorescence peak wavelength is defined as the Stokes shift, while the relative efficiency of fluorescence is measured using the quantum yield parameter, defined as the ratio of the emitted divided by the absorbed radiation [187]. The two categories of known fluorescent materials regard bulk fluorescent materials like ruby crystal and nanofluorescent materials like the quantum dots, which consist of elements from the II–VI or III–V groups of the periodic table. Nanofluorescent materials demonstrate significant modularity of optical properties using surface chemistry processes and modification of size [198]. Quantum dots can be applied to surfaces as a mixture with monomers and polymerized through exposure to UV radiation [199]. Fluorescent materials demonstrate a substantial potential for urban cooling [187].

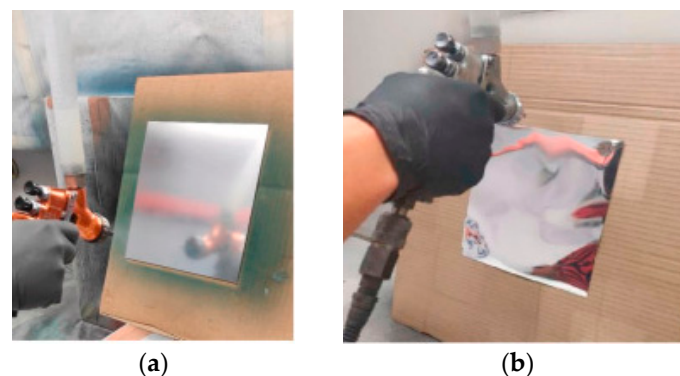
Implementing cool materials on a building's exterior surfaces is a passive and effective strategy to restrict the energy demand for cooling [200] and to enhance thermal comfort conditions during cooling demand periods [201]. However, cool materials belong to the category of seasonal envelope retrofit techniques because their energy impact is solely restricted during the summer period [202]. Specifically, the cooling and subcooling potential of cool materials, especially daytime cooling materials, is considered a severe drawback during the heating season. For instance, Khan et al. [201] examined the application of three different supercool building coatings on a city scale at Kolkata, India, categorized by high total reflectance, equal to 90% and 96%, and high total emittance, equal to 97%. According to their calculations, the maximum city-scale temperature decrease can be equal to 2.8  $^{\circ}\text{C}$  and 1.5  $^{\circ}\text{C}$ , at day- and nighttime, respectively, while the maximum possible heating penalty is calculated as equal to 186  $\text{W}/\text{m}^2$ . However, heating penalties induced by the overcooling phenomenon can be counterbalanced by the induced energy savings for cooling. More specifically, in a study by Xu et al. [203], the application of cool and supercool materials on building rooftops and façades is investigated regarding various residential neighborhood morphologies for the city of Nanjing in China. According to their calculations, cool and supercool materials are calculated to enhance buildings' energy performance during the cooling period by up to 13.85% and 17.74%, respectively, while the annual energy savings are found to be restricted to 1.95% and 2.35%, respectively. Following this, it is documented that for the Mediterranean climatic conditions of Greece, high-reflectivity roof coatings decrease incoming heat gains by around 17%, in comparison with low-reflectivity roof coatings, while restricting the peak cooling load to 11% less for the case of air-conditioned buildings [204]. According to Carlosena et al. [205], highly emissive radiative cool coatings are suitable renovation techniques for warm climatic conditions, especially for locations with prolonged summer periods.

Additionally, the combinational use of cool materials with conventional energy efficiency retrofits can result in the maximization of cool materials' contribution to the building's energy performance. For instance, Vakilinezhad and Khabir report that the opti-

imum performance of cool coatings is a function of climatic conditions and the insulation thickness of the construction element on which they are applied [202]. Suhendri et al. [206] examined the combined application of solar chimneys with radiative cool coatings and concluded that for the climatic conditions of Athens, this retrofit strategy can result in extended nighttime natural ventilation of up to 3 °C cooler ambient air. Moreover, in a study by Cheng et al. [207], the combination of phase change materials and cool coatings is highlighted as an effective strategy for the enhancement of indoor thermal comfort in buildings for the climatic conditions of Hefei, China.

Except for the regulation of cooling thermal loads in buildings, the collateral advantages of cool coating applications on buildings involve the enhancement of the neighboring urban climate due to the lower building surface temperatures and air temperatures [6]. The avoidance of traditional heat-absorbing material in the urban construction surface, including building rooftops and public outer spaces, can alleviate the urban heat island phenomenon that strikes cities on a worldwide scale [204]. For instance, according to a study by Zhu et al. [208] in various Chinese cities, the integration of cool coatings and greenery in urban public spaces and building rooftops and façades can result in a cumulative decrease in the ambient air of around 0.25 °C to 0.4 °C. Lastly, according to a city-scale simulation study of the city of Bandung, Indonesia, the deployment of greenery and cool coating on buildings' façades and rooftops resulted in an average decrease in air temperature of 0.65 °C and 0.53 °C, respectively.

Regarding the commercializing of low-cost cool materials, Carlosena et al. [209] developed and tested various daytime cooling materials with the use of aluminum or Vikuiti substrates and silica-based sprayed emissive coatings appropriate for building applications. The spray deposition of emissive coating onto two different substrates is illustrated in Figure 13. The developed daytime cooling materials are characterized by a total reflectivity of 70% and emissivity at the atmospheric window of 34% for the case of the aluminum substrate and by a total reflectivity of 97% and emissivity at the atmospheric windows of 89% for the case of the Vikuiti substrate. The recorded surface temperature is on average 1.70 K and 2.70 K lower than the ambient temperature for the cooling materials with aluminum and Vikuiti substrates with sprayed emissive coatings. The findings of this experimental study indicate that sprayed emissive coatings can be an economic and efficient solution for building applications with concrete, glass, and ceramic as the dominant substrates. However, it is of vital importance that optical degradation and overcooling during winter periods be moderated for cool materials to be an appropriate solution for a wide range of climatological conditions for a long lifespan.



**Figure 13.** Application of emissive coating with spray deposition onto (a) metallic substrate, and (b) plastic substrate (data were retrieved from Ref. [209]).

## 5.2. Cooling Materials of Dynamic Optical Properties

Thermochromic materials are innovative, active, dynamic, and adaptive materials that modulate their color as a function of external stimuli, for instance, temperature, solar irradiation, chemicals, or mechanical impacts [7]. Thermochromism is realized with

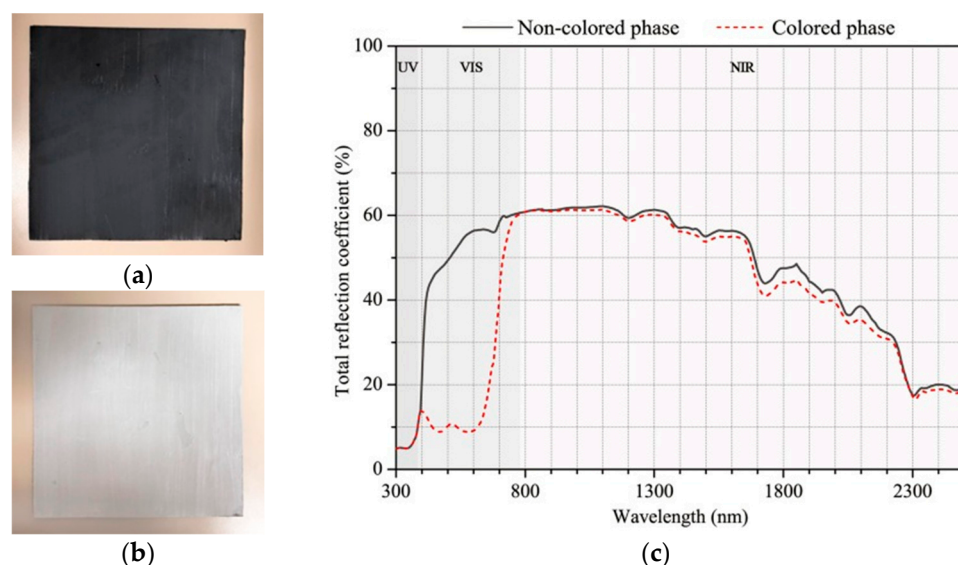
several mechanisms, for instance, the modulation of light in sol–gel films, surface plasmon absorption, phase transition in liquid crystals [210], aggregation and disaggregation mechanisms between dye and dye and dye polymers, modulation of the refractive index in photonic crystals [211], and variation of pH with temperature [7]. According to Garshasbi and Santamouris [7], thermochromic materials are categorized into two sub-groups, dye-based and non-dye-based. Dye-based thermochromic materials, and specifically, leuco dyes, are widely, and commercially available [212] and have been applied to and tested on buildings [213,214].

Leuco dye-based thermochromic materials are a mixture of common white paint in which the thermochromic pigments are incorporated [215,216]. The thermochromic pigments are three-component systems, encapsulated into polymer micro-shells [217,218]. More specifically, the thermochromic pigments are composed of the color former or leuco dye (usually cyclic ester) that functions as an electron donor and determines the color of the paint in the colored state, the color developer (usually weak acid or phenol [219]) that functions as an electron receiver and is responsible for the reversible color-change capacity of the pigment, and the solvent (usually long-chain alkyl alcohol or ester or acid) whose melting point and its transition to the liquid phase determines the transition temperature of the pigment, by which the entire reversible color-change procedure is controlled [213]. If the temperature of the thermochromic mixture is below the transition temperature of the solvent, the dye and the color former are bonded (zwitterionic structure), and the absorption wavelength window of the pigment encompasses the visible range and the dye is colored. Conversely, the increase in the pigment's temperature over the transition temperature of the solvent leads to the liquefaction of the solvent and the breakdown of the bond between the dye and the former color. The absorption wavelength window of the pigment does not overcome the ultraviolet wavelength range, and therefore, all the visible wavelengths of light are reflected, resulting in the colorless or white state of the dye [200]. Figure 14 illustrates a dye-based black-colored thermochromic paint for building applications.

Therefore, the variable wavelength absorption window of the thermochromic materials allows them to switch between a colored and a colorless state. These two states extend over different temperature ranges and are characterized by different optical properties. More analytically, in the colored state, the material is characterized by a low value of reflectivity, whereas in the colorless state, the material presents a high value of reflectivity [220,221]. For the thermochromic dyes to be utilized in the building environment, they should be incorporated into building materials, for example, in cement [212]. For the production of coatings, the encapsulated thermochromic dye-based materials are compounded with binders that function as polymer substrates allowing the film-formation of the coating, solvents that control the viscosity of the final product, pigments, dyes, and fillers, which induce color and special characteristics to the coatings and various additives, such as ultraviolet-stabilizers, and wetting and dispersing agents [212].

Multiple thermochromic coatings have been developed, measured, and tested after being exposed to sunlight for a few days or hours, or to accelerated climatic conditions in ultraviolet chambers. Ma et al. [216] developed leuco dye-based thermochromic materials that exhibited different solar absorptance values below and over the temperature of 20 °C, and a 4 °C lower surface temperature in their colorless state in comparison with a common-colored coating. In 2009, Karlessi et al. [213] created eleven thermochromic coatings with a transition temperature of 30 °C that either contained titanium dioxide, TiO<sub>2</sub>, or not. They also developed a cool coating, with and without TiO<sub>2</sub> and common coatings, to conduct comparative measurements. The developed coatings were studied for ten days in August–September, and it was observed that the surface temperature of the thermochromic coatings was lower in comparison with the cool and common coatings of a similar color. In addition, the TiO<sub>2</sub> additive resulted in cooler surface temperatures, whereas all thermochromic coatings presented high reflectivity in the near-infrared spectrum. However, at the end of the experiment, the solar reflectivity

difference between the colored and colorless state was decreased because of the bleaching of the colored state and the darkening of the colorless state.



**Figure 14.** (a) Colored and (b) colorless state of thermochromic paint (50 wt% toluene–xylene-based solvent, 30 wt% petroleum-derived resins, and 20 wt% thermochromic black pigments) sprayed over the polyurethane membrane, and (c) comparison of spectral reflectivity of the two states (data were retrieved from Ref. [222]).

In 2015, Zheng et al. [223] developed eight thermochromic coatings with a transition temperature of 25 °C and concluded that the titanium dioxide additives augment the sunlight reflectance importantly, while the dye content has a poor influence on the thermochromic coating's optical properties. They concluded that the sample with 5 wt% thermochromic dye and 10 wt% TiO<sub>2</sub> leads to the most energy-efficient coating. Moreover, in 2019, Zhang et al. [224] developed twelve thermochromic coatings with a transition temperature of 31 °C to examine the influence of the thermochromic coating content in thermochromic powder and TiO<sub>2</sub> additives as well as the influence of the TiO<sub>2</sub> particle size on the coatings' optical properties. They concluded that the presence of TiO<sub>2</sub> increases the coatings' reflectance, while the content of thermochromic powder presented no effect on their optical properties. The maximum difference in the solar reflectance between colored and colorless states was observed for the TiO<sub>2</sub> particle diameter of 60 nm.

Photodegradation, aging, or photobleaching are crucial disadvantages of the organic compounds of dye-based thermochromic materials that diminish their optical superior properties by a short period of a few months, hindering their outdoor application [225]. It is reported that ultraviolet light ignited an irreversible photochemical reaction which results in the reduction in molecular oxygen and the generation of reactive oxygen species [226]. However, according to a study by Karlessi and Santamouris [227], not only exposure to ultraviolet radiation but also to other ranges of the solar spectrum results in the aging of thermochromic dye-based materials and the corruption of their chemical structure. The combinational use of ultraviolet and red filters is the most effective at stabilizing the material's optical properties through time. Other researchers have also examined the integration of phase change materials in conjunction with thermochromic pigments on cement plasters as a solution against photodegradation [228]. Another crucial parameter of the natural aging process of optically advanced coatings is the accumulation of dust particles on their surface [229]. Specifically, according to the natural aging test conducted by Lei et al. [230], the shortwave reflectivity of highly reflective materials is reported to be reduced by 14% during two months. Additionally, another study by Di Giuseppe et al. [231] focused on investigating the effect of outdoor exposure on the angular reflectivity indexes of retro-reflective and diffusive coatings and concluded that the global reflectivity



index deteriorates for both material categories, whereas soiling phenomena and aging are reported to effect the diffusive coatings less.

Recently, in 2022, Wang et al. [232] developed a temperature-modulating radiative cooling coating for building applications, using a radiative porous membrane, of high solar and mid-infrared reflectance and high emittance at the atmospheric window, as a substrate and a red dye-based thermochromic top layer with a transition temperature of 31 °C. The temperature-induced color-change procedure presents the thermal hysteresis effect [211], which is longer during the cooling of the coating and its transition to the colored state, with a maximum width of 16 °C. To avoid the yellowing of the thermochromic pigments' appearance due to exposure to sunlight, a light stabilizer was used as an additive, ultraviolet-absorber, top layer to the coating. The final product was tested in an accelerating lighting chamber for 50 h, and no yellowing in the coating's appearance was observed, while after five color transition cycles, the total reflectance of the product remained unchanged. Their final product presents a high and low reflectivity value of 91.25% and 72.71%.

Regarding the integration of thermochromic materials in the urban structure and the improvement in the urban microclimate, Zhang et al. [233] examined the incorporation of dye-based thermochromic material in pavements for the city of Shanghai. In contrast to the conventional asphalt pavement, thermoregulating pavements of red and black-colored thermochromic pigments can achieve a temperature reduction of up to 10.60 K. Moreover, thermoregulating pavements outperform the respective cool-colored pavements by around a 2.0 to 5.0 K temperature difference. According to Fabiani et al. [222], thermochromic temperature thresholds of around 30.0 °C are more effective for heat-mitigating purposes. In this direction, Liu et al. [234] examined twelve color pigments most commonly found in building applications regarding the difference in reflectance between the colored and colorless phases of the thermochromic material. They found that the use of dark green and blue pigments in thermochromic coatings with a transition temperature of 33.0 °C demonstrated the highest changes in total reflectance, equal to 18.23% and 17.90%, respectively.

Integrating thermochromic materials into buildings is an efficient retrofit technique that can decrease energy demand throughout the entire year [235]. In a recent study, the application of thermochromic coatings on building rooftops was parametrically investigated regarding the total thermal transmittance of the rooftop construction and the coating's mean transition temperature and transition temperature range [236]. Thermochromic coatings demonstrated more efficient thermal behavior for the climatic conditions of Athens, resulting in yearly energy savings between the range of 2.2% and 17.7%, while the appropriate selection of optical transition parameters restricts heating penalties to less than 2.0%. However, a study that examined the impact on indoor thermal comfort conditions highlighted that thermochromic rooftop coatings, despite the reduction in the yearly energy consumption for both heating and cooling, can increase thermal dissatisfaction indicators, a phenomenon which is linked to the relatively slow response of the rooftop thermal mass [237].

Following this, Butt et al. [238] examined the impact of thermochromic coating, characterized by a low and high total reflectance of 0.20 and 0.80, on building energy performance, through a parametric analysis of three different building topologies, including a residential, an office, and a tertiary building, for the climatic conditions of Amsterdam, Seville, and Milan. The Spanish climatic conditions are more suitable for temperature-induced color adaptive coatings, resulting in up to 13.0% and 19.0% annual energy savings for the case of the tertiary building and residence, respectively. Following this, Zinzi et al. [186] studied the energy impact of thermochromic rooftops on the energy performance of highly and low-insulated one-story residential buildings, regarding the Mediterranean climatic conditions of Barcelona, Palermo, and Cairo. The examined thermochromic coatings, characterized by a maximum total reflectance switch of 0.60, are found to demonstrate an annual reduction in energy consumption that reaches up to 8.5% and 19.0% in comparison to cool-colored and high-absorptive roofs, respectively. In a study by Berardi et al. [239], the energy-saving potential of four different thermochromic façade and roof coatings with various integrated



color pigments is investigated for a four-story office building in Toronto. The coating is characterized by a phase transition temperature of 31.0 °C and is found to result in up to an 8.9% reduction in the cooling energy demand and a restricted overcooling penalty of 1.7%.

### 5.3. Trends and Challenges of Optically Advanced Coatings with Cooling Potential

The application of a cool-colored coating on a building's exterior is a common energy-efficient action that can result in significant cooling energy savings at the building and city scale. Additionally, the use of dynamically adapted cool-colored material can significantly restrict heating penalties. However, the extensive and durable application of this technology as a heat mitigation and energy-saving solution demands further research on two main domains. The first one concerns the combat of the degradation phenomenon of leuco dye thermochromic materials through the development of suitable nano-encapsulation structures to isolate the materials from the environmental conditions [183]. Except for the optical spectral filters reviewed previously, no further technological solutions have been developed yet. The second domain concerns the analysis of non-leuco dye thermochromic materials, characterized by higher optical stability, for instance, liquid crystals, photonic crystals, quantum dots, and plasmonic materials [187]. In this direction, the use of inorganic vanadium dioxide in opaque building components, for instance, tiles [240] or concrete [241], is an alternative solution, the effectiveness and suitability of which demand further research and testing.

## 6. Mechanical Ventilation and Bioclimatic Design Technologies

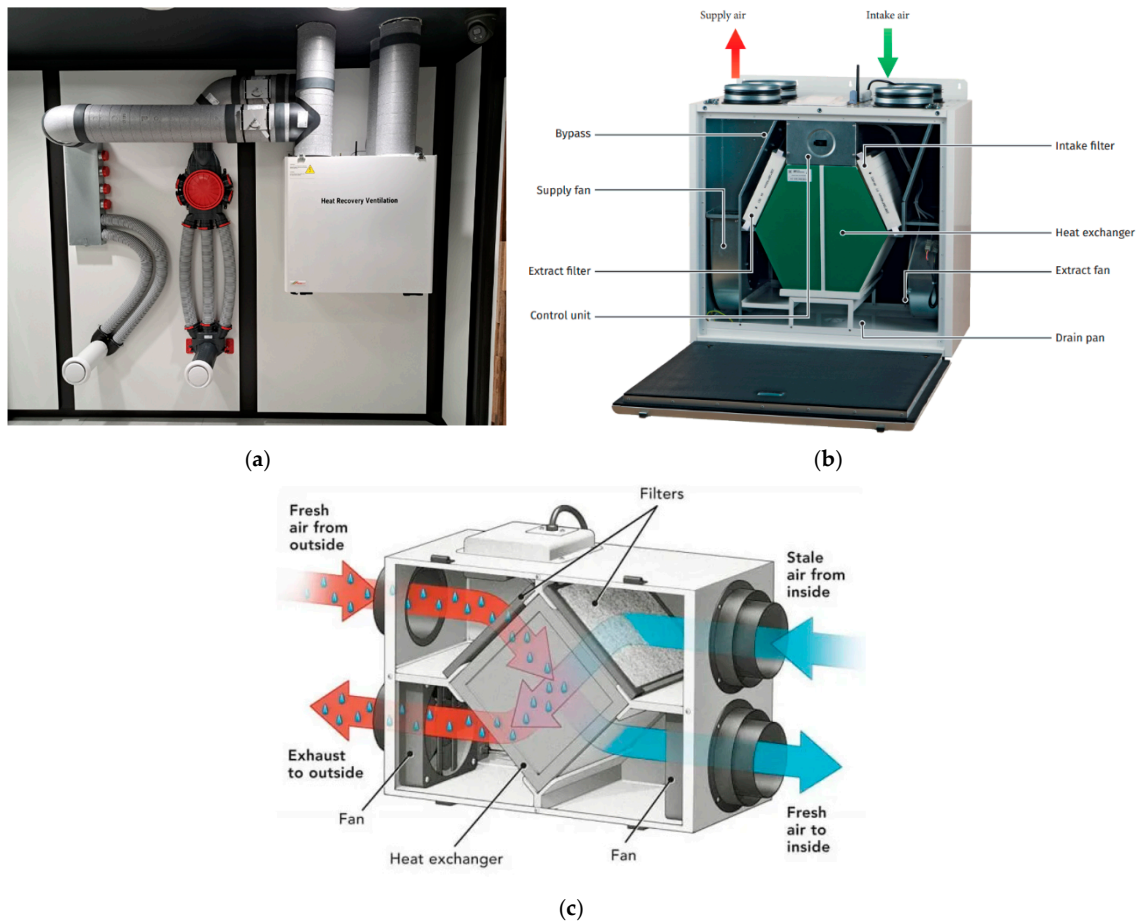
In this final section, an analytical discussion regarding the passive and bioclimatic building envelope technologies for sustainable design is conducted. Specifically, the analyzed energy-efficient solutions concern the system of mechanical ventilation with heat recovery, the use of shading elements, the bioclimatic design of solar chimneys, and the infrastructures of green façades and green roofs. These passive technological solutions can greatly affect the thermal behavior of buildings, offering building-scale and even city-scale advantages regarding energy savings and thermal comfort.

### 6.1. Mechanical Ventilation

The installation of mechanical ventilation systems (MVSs) is of primary importance for the adequate, controlled, and efficient air renewal of highly efficient buildings. An important feature in the design of highly efficient buildings is the high airtightness of their thermal envelope [242]. More specifically, the restriction of thermal losses due to infiltration significantly affects the building's energy performance and operation of its energy systems, the indoor thermal comfort conditions as well as the indoor air quality [243]. According to the existing literature, thermal losses due to infiltration account for 15.0% up to 50.0% of the energy consumption for heating and cooling purposes in residential [244] or industrial buildings [245]. Highly airtight buildings require the installation of MVSs to ensure sufficient air renewal. Mechanical ventilation is defined as the active process of delivering and extracting air from an indoor space, powered by electrically driven equipment, for instance, motor-driven fans and blowers [246]. Mechanical ventilation systems are distinct from air handling units or compact heating, cooling, and ventilation units, and can only partially cover a building's heating and cooling through the incorporation of heat recovery components. The mechanical ventilation system with an incorporated heat recovery unit is an energy-efficient solution that only concerns highly efficient, airtight buildings.

Heat recovery is a common solution that can be applied in conjunction with mechanical ventilation systems in buildings and is increasingly used to decrease the energy demand for space cooling and heating. During the heat recovery process, the exhaust air is utilized as either a heat source or heat sink depending on the ambient conditions, season, and energy requirements of the building [247]. More specifically, the thermal energy in the exhaust air is transferred to the incoming fresh air, resulting in raising its temperature and decreasing the energy demand during the heating season. Similarly, in the cooling

season, the exhaust air is exploited as a heat sink, in which the warmer incoming air extracts its thermal energy resulting in the reduction in the building's energy demand for cooling. No humidity transaction is conducted between the two streams. Mechanical ventilation with heat recovery is characterized by a nominal efficiency in the range of 70.0% to 95.0% [248]. Figure 15 illustrates an MVS and its components regarding heat recovery.



**Figure 15.** Ventilation with heat recovery. (a) System and ductwork, (b) component diagram, and (c) schematic of heat exchange between exhaust and supply flows (data were retrieved from Ref. [249]).

Mechanical ventilation with heat recovery can result in a decrease in energy consumption for both space heating and cooling. More specifically, according to a study by Bellos et al. [250], in which the holistic energy renovation of a multi-family building in Athens is investigated, mechanical ventilation units with heat recovery, external wall insulation, and triple-glazed windows are the most energy-efficient techniques, which result in combinational energy savings for heating and cooling that are equal to 93.0% and 78.0%, respectively. In another study, mechanical ventilation was examined as a cooling retrofit action for a residential building for the four climatic categories of Greece [163]. As a standalone action, mechanical ventilation is calculated to result in cooling energy savings of 10.1–17.5%, while when combined with cool rooftop coating, cooling energy savings are increased by 32.1–47.1%. Additionally, Grygierek and Grygierek [251] examined various ventilation strategies in a multi-family building in Poland and concluded that natural ventilation through windows can increase heating energy demand by up to eight times, compromising the indoor thermal comfort conditions. On the other hand, the use of mechanical ventilation with heat recovery is recorded to induce instantaneous load energy savings of up to 50.0%. Following this, in a case study of a two-story residential building in Changzhou, China, a location characterized by cold winters and hot summers,

the standalone installation of a ventilation system with a heat recovery system resulted in 14.0% energy savings for cooling [252]. In a study by Karaiskos et al. [253], natural ventilation and mechanical ventilation with heat recovery are compared in terms of indoor air quality and thermal conditions for the case of a tiny building in San Antonio, Texas. According to their results, mechanical ventilation with heat recovery ensures better indoor air conditions. However, the low airtightness of the building, around 5.4 air changes per hour, is the main obstacle to the efficient operation of MVSs, and its high superiority against natural ventilation is the specific case study.

Secondary applications of mechanical ventilation regard ventilative cooling, used as a cooling technique, and dilutive ventilation, which is necessary to provide high air quality and combat airborne transmission, especially in public buildings of high occupancy [254]. The proper design of the operation of the MVS can contribute to the reduction in building cooling loads and cumulative energy consumption for cooling, providing ventilative cooling [255]. For mechanical ventilative cooling to be energy efficient, the optimization of the system's ventilation rate and the use of high-efficiency fans are required [256]. The efficient operation of electrically driven fans in MVSs secures low energy consumption of mechanical ventilation and the extension of the time that ventilative cooling can be used without overly increasing the building's total electricity consumption. Two factors that hinder the efficient operation of MVSs concern the installation of centralized systems, which is combined with high ventilation duct resistance, mainly for the case of mid- or high-rise buildings [255], and the dust accumulation on the system filters [257].

Ventilative cooling can be applied in the form of night ventilation, which exploits the drop in the night ambient temperature by inserting cool air into the building and removing the heat that accumulates during the day [258]. In a study by Zhang et al. [258], the optimization of the MVS is investigated for the night operation of an office room in Beijing. They reported that energy cooling savings mainly derive from the load-shifting effect and heat extraction from building materials. According to their calculations, the optimized mechanical ventilation rate can reach up to 17.0 air changes per hour and result in 47.0% energy saving during the summer period.

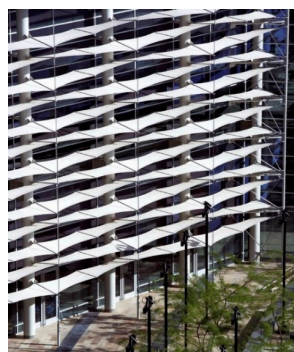
According to Sha and Qi [255], for applications in high-rise buildings, the control of the MVS should take into consideration the energy efficiency of the ventilation fans to determine the optimal air flow rate. Based on their study, mechanical ventilation can result in 43.0% cooling energy savings, regarding the climatic conditions of Canada and Northern Europe. Following this, Motuzienė et al. [259] underlined that the operation of MVSs according to values that were set during the design stage is the primary reason for the performance gap observed in office buildings. According to monitoring and simulation analysis of an office building located in Vilnius, Lithuania, they calculated that controlling the operation of the ventilation rate based on realistic occupancy schedules can result in a 30.0% reduction in electricity consumption. Finally, in a study by Maask et al. [260], MVSs are studied in terms of flexibility-side demand, according to which electricity consumption is adapted to external stimuli, for instance, electricity availability, due to the integration of intermittent renewable energy sources, electricity price, or government incentives. Their control modeling concerns the forecast of the system's capacity, duration, and pricing, with the simultaneous monitoring of indoor air quality and thermal parameters, namely temperature, humidity, and carbon dioxide concentration. This method is applied for the cases of constant and variable air flow rate systems resulting in a 0.8% and 6.5% forecasting error and a 4.0% and 1.1% reduced cost in electricity, respectively.

## 6.2. External Shading

Solar shading components and design are of pivotal importance for effective building energy performance, contributing to the management of solar heat gains, energy consumption, and thermal and visual comfort conditions. Therefore, shading design needs to be a main retrofitting approach or be included in the early design process for new constructions, especially when referring to highly glazed buildings [261].



External shading configurations are mainly categorized as fixed or dynamic. Fixed shading is currently the predominant commercially available solution in the building sector [262], and based on the geometry of shading components, they can be discerned into horizontal, vertical, or egg-crate-shaped configurations, as depicted in Figure 16a–c, and when applied extensively onto a façade, they can create double-skin perforated shades [263]. In contrast to fixed shading, dynamic shading configurations can adapt to variable weather conditions and adjust to optimal shading positions. Examples of dynamic shading configurations include outdoor blinds [264], overhangs [265], roll shades [266], awnings, or more complex designs such as origami-based dynamic facades [267], as shown in Figure 16d. Dynamic shading components are innovative and intelligent systems that automatically operate to preserve the desired indoor conditions with simultaneous energy conservation, according to external stimuli [268]. The domain of dynamic shading devices and their control system has been widely investigated, and research has focused on the employment of weather, indoor environment, and buildings' energy system performance data as inputs of their control systems [269]. The concept of dynamic shading lies in the upgrade of conventional shading solutions to the achievement of higher energy efficiency, better visual and thermal comfort, and lower energy consumption for space cooling and lighting. Figure 17a,b illustrate a vertical and a horizontal adjustable photovoltaic louver system. The selection of the appropriate shading system is determined by the building location, orientation, and climatic conditions, and the architectural characteristics of the façade onto which it is implemented. Additionally, economic parameters linked with the maintenance cost or the complexity of the control system, as in the case of dynamic shading, play an important role in the decision as to the most appropriate shading strategy for a building [270].



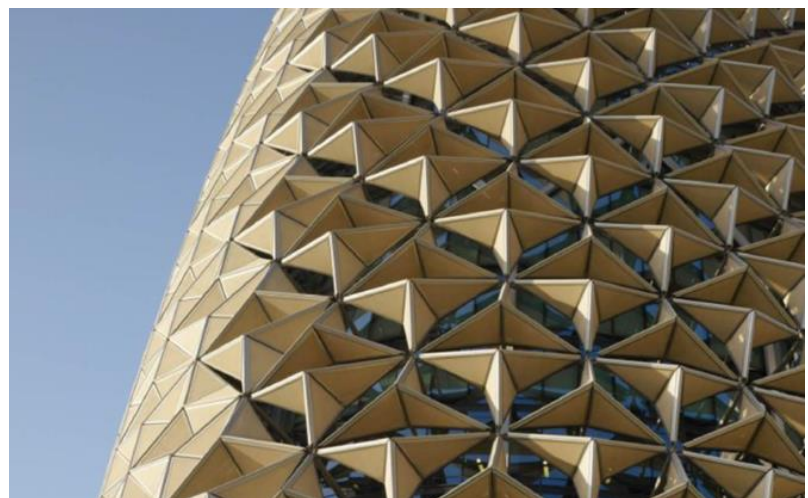
(a)



(b)



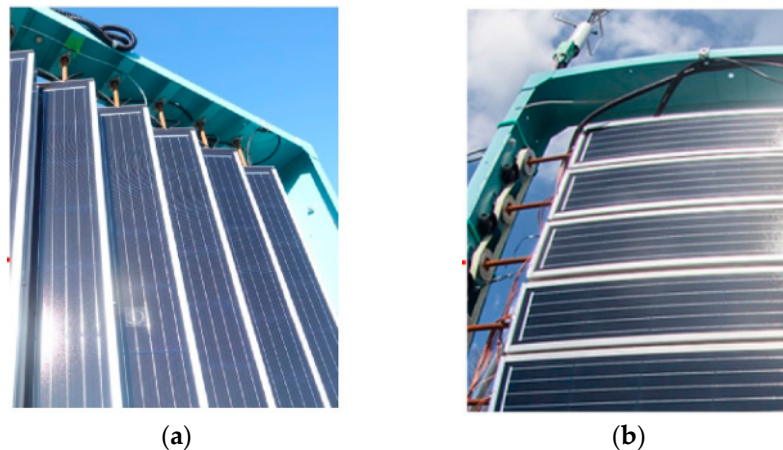
(c)



(d)

**Figure 16.** Fixed shading configurations of (a) horizontal, (b) vertical, and (c) egg-crate-shaped. (d) Origami-based dynamic (data were retrieved from Refs. [271–274]).

Low maintenance cost is the primary reason that supports the widespread application of static shading systems [275]. In this direction, Méndez et al. [276] investigated the application of a novel static shading component that can result in an annually optimized building energy performance regarding hot climatic conditions. According to their calculations, static shading can alleviate thermal discomfort in non-air-conditioned residencies by up to 71.3% and reduce energy consumption for space cooling by up to 30.4%. The deployment of static shading techniques on the opaque structure of buildings is parametrically investigated and combined with the predominant cooling technique of cool paints, for multiple climatic categories in a study by [263]. According to their results, façade shading can result in 8.0% up to 28.0% cooling energy savings, while the combinational use of shading and cool paints can result in 10.0% up to 40.0% cooling energy savings. Another approach to shading refers to self-shading, and the integration of more complex building geometrical designs [262] or components that disrupt the continuity of the thermal envelope, such as balconies [277]. Despite the energy savings due to shading, this approach increases the thermal bridge effect of buildings. Thorough energy analysis is necessary to optimize thermal management [278] and to calculate the indirect thermal losses induced by geometrical thermal bridges assessing the impact on the building's energy performance during the entire year.



**Figure 17.** (a) Vertical and (b) horizontal adjustable photovoltaic louver system (data were retrieved from Ref. [279]).

The conventional materials of shading components include concrete, timber, steel, and aluminum [263]. However, alternative solutions regarding the material and composition of shading components have been integrated into the building sector. For instance, in a study by Garcia-Nevado et al. [280], high-mounted shading configurations of textiles, specifically sun sails, are examined as a slightly invasive cooling technique in the city of Cordoba, Spain. According to the thermography results, sun sails are found to result in temperature decreases in both the ground and building façades, equal to 16 K and 6 K, respectively. Additionally, the use of photovoltaic panels is reported to offer indirect shading benefits to building constructions. The integration of photovoltaic panels can be either stationary, in the form of building-integrated photovoltaic panels [281], or dynamic. More specifically, Jiang et al. [282] investigated the integration of static and dynamic photovoltaic shading devices into a room of an office building in the heating-dominated city of Qingdao, China, to examine the effect on the building energy demand and daylight illuminance levels. The examined control strategies involve the rotary movement of the shading configuration, the sliding movement, and a hybrid movement that changes both the tilt angle and installation height of the device. For each strategy, the optimum monthly position is defined. According to their analysis results, static photovoltaic hybrid devices can result in yearly energy savings that reach up to 40.6% in the case of a 35° tilt angle. For

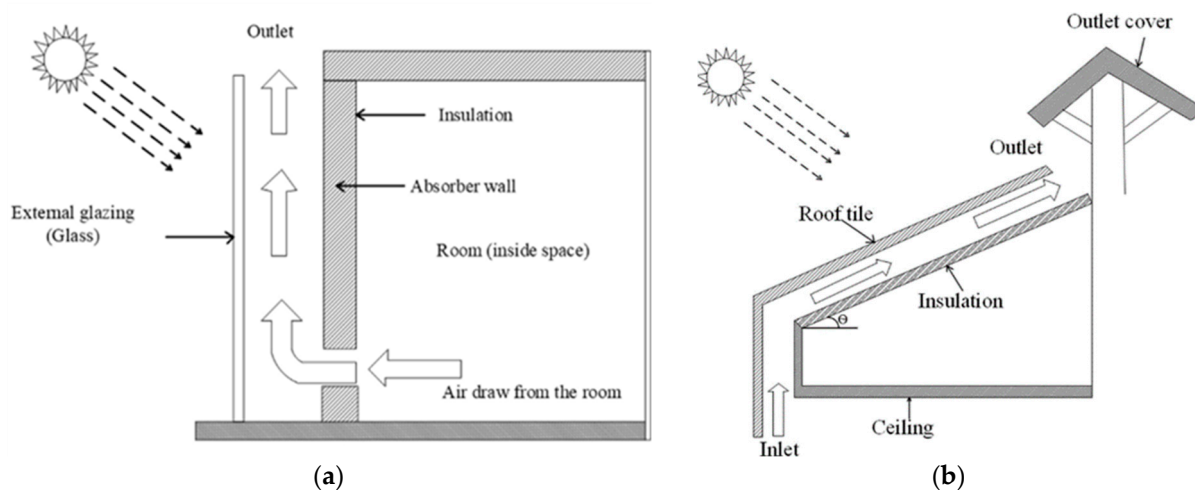


the rotation, the sliding, and the hybrid strategy, the yearly energy savings are equal to 31.1%, 47.2%, and 50.4%, respectively.

Following this, Krarti [265] investigated the integration of dynamic photovoltaic overhangs on residential buildings for the US climatic conditions and calculated a decrease in the total energy demand that reaches up to 90.0%. In this direction, Lee et al. [283] developed a light shelf, or else a hinge structure applied onto windows with integrated photovoltaic modules, that dynamically adjusts the module inclination angle by a folding movement, with a minimum intervention to the indoor daylight illuminance. Following this, Wu and Zhang [284] investigated the implementation of origami-based dynamic shading in a small office building in a location of China characterized by hot summers and cold winters. For the examined building with high glazing area, dynamic shading is found to enhance daylighting and energy performance by 91.5% and 19.9%, respectively. On the other hand, for the cooling-dominant climate of Tehran, the integration of shading devices in a small office building can decrease energy consumption by up to 29.0%, with a simultaneous increase in thermal comfort conditions of up to 56.7%.

### 6.3. Solar Chimney

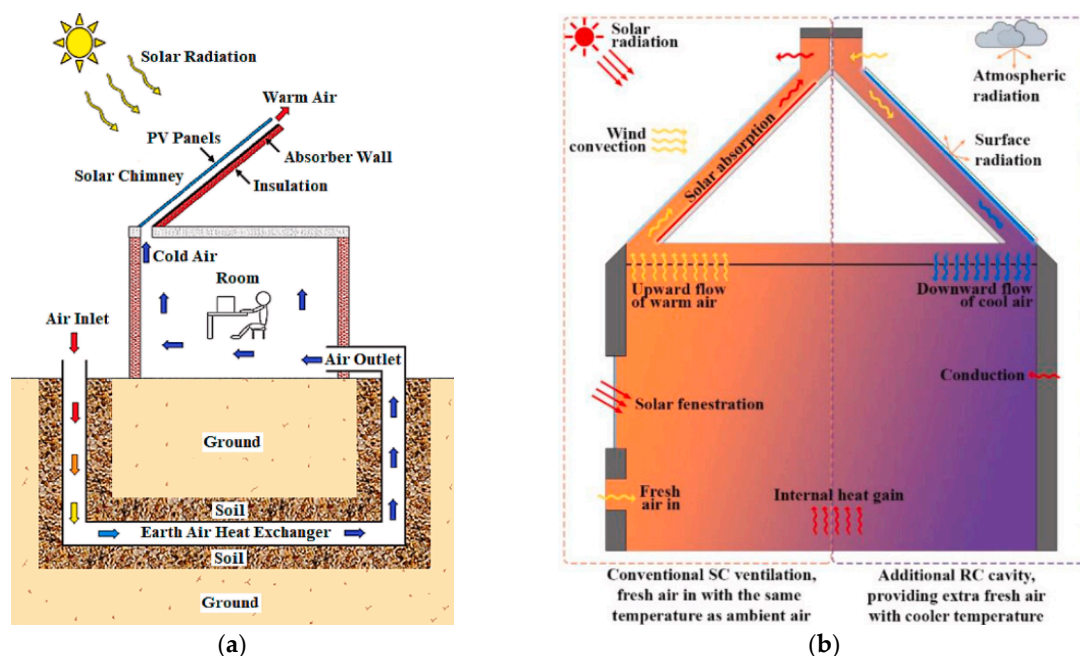
The solar chimney is a simple architectural and bioclimatic design that utilizes solar irradiation to enhance natural ventilation [285]. The solar chimney is an air channel, composed of an absorptive structural element with high heat capacity and a glazing system characterized by high transmittance, and exploits the mechanism of thermal buoyancy to induce natural ventilation in buildings. More specifically, air within the air channel is heated through convection, which causes a temperature increase and consequent decrease in its density. The heated air rises and is removed to the ambient environment, while air from the interior building space fills the space created and cooler air is drawn into the building in a continuous cycle [286]. It is reported that in some cases, photovoltaic cells are integrated into the glazing system, or the latter can be substituted by an opaque cover. The solar chimney can be incorporated into either a wall, vertical design, or a roof, inclined design, as depicted in Figure 18 [287].



**Figure 18.** Schematic of the solar chimney incorporated into (a) wall and (b) roof (data were retrieved from Ref. [287]).

A solar chimney can be combined with various active or passive technological solutions aiming to enhance its contribution to the building's thermal comfort conditions and energy balance. For instance, Alkaragoly et al. [288] investigated a passive–active combinational system composed of a solar chimney, a photovoltaic system, and an earth–air heat exchanger, to improve natural ventilation and thermal comfort conditions, and simultaneously exploit the produced electricity to cover the energy need for space cooling. The aforementioned system is illustrated in Figure 19a. Following this, Suhendri et al. [197]

examined a conventional south-facing solar chimney roof combined with a sun-opposite cavity covered with cooling material, as depicted in Figure 19b. The additional radiative cavity is found to increase the building ventilation rate by up to 2.1 air changes per hour and reduce the room temperature by up to 2 K, in the case of dryer climatic conditions. Another solution refers to the combination of solar chimneys with phase change materials. In this direction, Nateghi and Jahangir examined the impact of solar chimneys coupled with phase change material on the building's thermal comfort for various climatic categories. According to their calculations, a PCM-doped solar chimney is effective for high daily temperature fluctuations. Additionally, the examined system demonstrated the most efficient performance during the cooling period contribution for Tehran, a city characterized by a cold semi-arid climate, and the worst during both heating and cooling periods for the hot-arid climate of Yazd. Beyond the scope of energy efficiency, solar chimneys can be exploited to enhance indoor air quality. Therefore, Li et al. [289] examined the combination of a small-scale solar chimney with various configurations of photocatalytic reactors to achieve methane degradation.



**Figure 19.** Schematic of solar chimney coupled with (a) photovoltaics and an earth–air heat exchanger, (b) radiative cooling cavity (data were retrieved from Refs. [288,290]).

#### 6.4. Green Façades and Green Roof Systems

The thermal and optical properties of the materials used in urban structures determine the amount of heat absorbed, reflected, emitted, stored, and dissipated through convection and evaporation. Green façades and roofs are discerned as an alternative solution to the urban infrastructure, characterized by multiple building-scale and city-scale advantages, as well as advantages regarding social health, quality of life, and psychology [291]. More specifically, the global market size of green roofs and green façades reached USD 2.1 billion in 2023 and USD 662 million in 2020, respectively, exhibiting growth rates of 13.3% and 7.2% over the next decade [292]. Green walls, or green façade systems, are vertical green systems that can be classified into the two basic categories of living walls and green façades [293]. The latter regards the case where vegetation grows and expands directly on the building façade, covering it gradually, as depicted in Figure 20a, or the case where an additional support system is used, as depicted in Figure 20b [294]. On the other hand, living wall systems concern the use of materials and technologies that support the integration of a wide variety of plant species and the uniform coverage of large building surfaces, as illustrated in Figure 20c [293]. Following on from this, another form of greenery integration into the

building structure concerns green roof systems. Green roofs are categorized into either intensive or extensive systems [295]. Intensive green roofs, as depicted in Figure 20d, are characterized by a thick substrate, a wide plant variation that includes small trees, and a large weight as well as high maintenance and initial cost [296]. On the other hand, extensive green roof systems are light, low cost, and low maintenance, demanding configurations that can support only a thin and light layer of vegetation, typically grass [295].

The respective green infrastructure typologies can drastically elevate the urban scenery by increasing the green areas and achieving an adequate ecological reconciliation, leveraging air pollution [297] and managing the air temperature distribution [298], solar irradiation [299], and light scattering in cities. Except for the social benefits of reuniting citizens with nature and combating noise pollution, green façades and roofs are qualified as appropriate solutions for the mitigation of climate change and extreme heatwaves [300], and the improvement of the quality of life in urban environments [294].

Green roofs and façades operate as an additional layer over the building materials that protect them from thermal stress and prolong their durability. Additionally, they operate as an insulation layer that regulates heat losses through conduction and a solar absorbent that reflects only a small amount of the incident solar irradiation [301], minimizing light scattering and enhancing optical comfort conditions. Following on from this, the water content of green roofs and façades is responsible for these systems' high heat storage capacity and the mitigation of abrupt temperature variations on the building envelope surface and in the indoor building space. Lastly, the evaporation and transpiration processes of greenery result in a cooling effect of the surrounding environment and retention of low temperature levels at the building boundaries [302]. In this direction, green roofs and façade systems are most efficient when exploited in locations with hot and humid climatic conditions [303]. This is also supported by the study of Borràs et al. [304] who investigated the thermal behavior of a typical family house parametrically for the various climatic categories of Spain. They concluded that for higher annual temperatures and cumulative solar irradiation, as in the city of Almería, the refurbishment of the building rooftop to an extended green roof system can result in annual energy savings of 9.4%, which are comparable and higher than the energy savings achieved through the entire envelope's thermal improvement of 9.11%. An older but interesting study by Zinzi and Agnoli [305] compared the energy performance of green roofs and cool roofs for the climatic conditions of three south Mediterranean locations. They concluded that wet green roof systems outperform cool roofs in yearly energy savings by 24%, 11.6%, and 45.0% for Palermo, Barcelona, and Cairo, respectively. They highlighted that if dry conditions or water content given by actual rainfall is taken into consideration, the green roof systems are not the optimum energy renovation selection.

Nevertheless, green infrastructure solutions have been studied and adopted on a global scale and for multiple climatic conditions. More specifically, Wang et al. [306] investigate the energy-saving potential of green roofs when applied in the entire city of Xiamen in China. They report that the integration of greenery in the buildings' rooftops can result in total city-scale energy savings of up to 1.83% and peak load savings of up to 1.63% [306]. Regarding the topology of the respective city, green roofs are proven to be more energy-effective for low-rise residential and industrial buildings rather than high-rise public buildings. Following this, in a study by Báez-García et al. [307], a green façade system is numerically and experimentally tested regarding the thermal impact on heat transfer through the building's walls and the indoor air temperature, for the climatic conditions of the city of Cuernavaca, Mexico. According to their results, in comparison with the bare walls, green façades dynamically regulate heat exchange with the ambient environment, operating as an effective thermal barrier during the four seasons. During the most energy-demanding season of summer, the maximum recorded decrease in the indoor air temperature is calculated at 5.64 °C and in the heat flux at 16.79 W/m<sup>2</sup>.

Following this, Sharbafian et al. [308] examined the integration of a green wall façade system in a three-story residential building in Tehran. Their parametric analysis aims to

calculate the maximum heating and cooling loads of the building for various distances between the building's thermal envelope and the green wall façade's indirect system, as well as various densities of the greenery. According to their results, a less dense green wall façade system with a distance of 5 cm from the building wall results in a maximum decrease in the heating and cooling load of 15.1% and 17.9%, with a minor decrease in the building daylight autonomy that is restricted under 1.5% for both cases. The aforementioned system of an indirect green wall façade separated from the building envelope by a small distance can also be referred to as a double-skin green façade [309]. According to Convertino et al. [309], double-skin green façades enhance the building's energy equilibrium through the shading effect and the plant evapotranspiration process. The latter highlights double-skin green façades as a better cooling mitigation technique in contrast to common shading devices.

Except for the enhancement of indoor thermal conditions, green systems can result in better indoor air quality conditions. More specifically, in a study by Pappa et al. [310], the aerodynamic effect of green roof and façade systems on a building's natural ventilation, and therefore on the indoor air quality, is experimentally studied. According to their results, a green façade system can result in an up to 16% decrease in the building's air renewal rate, while a green roof system can enhance the overall natural ventilation rate, and therefore the indoor air quality, by 8%. However, green infrastructure solutions can have a negative impact on indoor thermal comfort conditions. For instance, a study by Nagdeve et al. [311] underscored the necessity of ventilation and humidity control systems in an experimental study integrating a green façade system in an office building in New Delhi, India.

A serious limitation to their extended use regards their high maintenance and initial cost, which are not easily counterbalanced by the heating and cooling energy savings of a building [312]. If restricted to the dimension of a building's energy performance, the enhancement induced by green walls and roofs can also be obtained through the adoption of more conventional renovation actions, such as thermal insulation materials, with less financial resources [312]. However, it is essential to emphasize the numerous benefits that green roofs can provide, particularly in the context of sustainable urban planning. Aspects such as urban heat island mitigation and hydrological management can become increasingly significant. Thus, energy savings should be regarded as just one of many factors, rather than the sole reason, for choosing this technology [313].

#### *6.5. Trends and Challenges of Mechanical Ventilation and Bioclimatic Design Technologies*

The incorporation of a mechanical ventilation system is a popular energy-efficient action when combined with a heat recovery unit. However, its application is only effective for the case of a highly airtight building with minimum thermal losses due to infiltration. Moreover, the incorporation of shading elements into the building envelope can result in important cooling energy saving, regulating solar heat gains and enhancing indoor thermal and optical comfort. Dynamic shading elements is an emerging solution that can properly adapt to environmental conditions and maximize the shading effect. However, their adoption demands high installation and maintenance costs due to its control system, making it a more expensive and unsuitable choice for low- or mid-rise residential buildings. Regarding the bioclimatic design of solar chimneys, the trend is to combine its use with energy-efficient technologies, such as thermal insulation, PCMs, low-e, and highly thermal resistive glazing. However, the thermal upgrade of a solar chimney transforms a simple passive solution into a multi-parameter, more expensive, and complicated system. Lastly, the solution of green infrastructure, namely green façades and green roofs, offers important building-scale and city-scale advantages. Their widespread adoption is a solution toward the design of sustainable cities and the combatting of the heat island phenomenon with a subsequent decrease in buildings' thermal loads.





**Figure 20.** (a) Traditional green façade, (b) green façade with a support system, (c) continuous living wall system, and (d) intensive green roof (data were retrieved from Refs. [293,314,315]).

## 7. Discussion

### 7.1. Comparative Analysis Regarding Energy Savings

The emerging industries of highly efficient thermal insulation materials concern vacuum-insulating panels (VIPs) and aerogels, the incorporation of which into the building envelope is recorded to reduce constructions' total thermal transmittance value by up to 88.0% [43]. For residential buildings, VIPs are found to result in energy savings of up to 68.7% for the winter period and up to 30.0% for the summer period. For larger tertiary buildings the maximum decrease in yearly energy consumption is 10.6%, but when combined with a ventilation system, it is 36.0%. VIPs and aerogels' small thickness requirement increases the available living space when the solution of internal insulation is selected. Aerogels are selected either because of their high thermal resistance (silica aerogel blankets) or because they combine advanced thermal and optical properties (semitransparent granular aerogel and brittle transparent monolithic aerogel). Aerogel-based materials outperform conventional thermal insulation of the same thermal resistance or radiative coatings of the same solar properties by up to 43.4% of extra cooling energy savings [56].

The use of multiple, low-conducting, glass panes combined with the use of transparent insulation in between the window panes, for instance, solid silica-based transparent materials, low-conducting materials, such as noble gases or vacuum, or the reduction in gas pressure within the window installation, can significantly improve windows' thermal performance [93]. Low-e coatings reduce radiation thermal losses by 1.5–3.0 W/(m<sup>2</sup>·K) for double-glazed windows and by 1.0–1.8 W/(m<sup>2</sup>·K) for triple-glazed windows [87]. Moreover, the incorporation of monolithic silica aerogel into low-e double-glazed windows can enhance yearly building energy savings by up to 55.0% [101]. Noble gases abate heat transfer through convection between window panes by around 10% (argon), while better performances can be recorded with the use of krypton or xenon [87]. Vacuum-

insulating windows, characterized by a total thermal transmittance value of less than  $1 \text{ W}/(\text{m}^2 \cdot \text{K})$  [108], can result in a 76.3% and 59.4% decrease in annual energy consumption, for cooling- and heating-dominated cities, respectively [112]. Lastly, chromogenic windows adapt to external stimuli [113]. Commercially available thermochromic glazing is found to induce yearly energy savings of 50.0% [119], while electrochromic glazing, which is less expensive, is found to induce savings of up to 18.5% [124].

Phase change materials are incorporated into buildings either as a material layer that when combined with thermal insulation can result in energy savings of 66.2% [169], or as a component of composite construction materials, for instance, cement [154], or ceramic-based components [154], mortar, foamed cement, and concrete [155]. The latter method ensures the retention of the material even after its complete condensation, the non-intake of moisture from the ambient air, and the decrease in cooling thermal loads by 54.0% for PCM-doped voided roof slabs [158], and 48.5% for PCM foamed concrete rooftop. Another solution that exploits a high thermal capacity mechanism is the Trombe wall, which is regularly combined with various technologies to achieve efficient thermal performance. A static PCM–Trombe wall configuration with radiative cooling properties is calculated to reduce overheating by 55.2% in contrast to the simple Trombe wall [177], while a dynamic PCM–Trombe wall is found to be 79.0% more thermally efficient than the static Trombe wall [179]. More complicated and expensive configurations of Trombe walls combined with innovative highly insulative window systems have also been investigated, resulting in comparative thermal enhancements of around 12.2% in contrast to typical Trombe walls [181].

Cool-colored and daytime radiative cooling materials are a passive and effective solution for the reduction in peak cooling load and total cooling energy demand by up to 11.0% and 17.7%, respectively [204], and the enhancement of indoor and outdoor thermal comfort conditions during cooling demand periods [201], especially for locations with prolonged summer periods [205]. Research advancements in the domain have led to the commercializing of low-cost cool materials that can be easily applied through the spray deposition method. Conversely, thermochromic coatings are an efficient, commercially available, year-round energy solution that can decrease annual energy demand by up to 19.0% in contrast to high absorptive rooftop coatings. Additionally, thermochromic coatings restrict overcooling penalties to less than 2.0% because of their temperature-modulating surface optical properties mechanism [236].

The installation of mechanical ventilation systems offers adequate, controlled, and efficient air renewal, which combined with heat recovery units can result in the reduction in energy demand for space heating and cooling by up to eight times in contrast to natural ventilation through windows. External static or dynamic shading components contribute to the management of solar heat gains, the decrease in energy consumption, and the enhancement of thermal and visual comfort conditions. Static shading is characterized by a low-maintenance cost and can alleviate thermal discomfort in non-air-conditioned residencies by up to 71.3%, and reduce energy consumption for space cooling by up to 30.4% [272]. Conversely, a dynamic shading system can effectively adapt to external weather conditions and result in higher annual energy savings of 50.4% [278]. Lastly, green roofs and façades operate as an additional layer over the building materials that protect them from thermal stress and prolong their durability, simultaneously managing indoor solar thermal gains and reducing the annual energy demand for heating and cooling by up to 45.0% for the case of wet green roofs [301].

## 7.2. Challenges and Future Work

VIPs and aerogels' share in the European thermal insulation market accounts for only 1%, which explains their high specific costs. Incorporating innovative insulation materials into the building envelope is an economically viable solution if savings in available living space and construction materials are taken into account. VIPs and aerogel-based materials are very fragile, demand specialized handling, and are non-receptive to on-site geometry

adjustments since the risk of perforation greatly diminishes their thermal properties, especially in the case of VIPs [36]. Parallely, the use of joints and supporting components for the installation of the panels induces thermal losses due to the creation of thermal bridges [39]. Despite their advanced thermal properties, the high specific production and installation costs diminish their comparative advantage against conventional insulation materials. The future of insulating materials depends on combining their great thermal properties, lightweight construction, low environmental impact, and easy installation. Therefore, research innovations aiming at increasing the lifespan of novel insulating materials of aerogels and VIPs and diminishing the risk of thermal degradation during installation are vital.

Innovative window systems that incorporate multiple panes or noble gases demand thicker, heavier, more expensive, and hermetically sealed glazing configurations. Vacuum-insulating windows are more advanced than the hermetically sealed configurations of gas-filled windows and require the presence of hermetic edge seals and mechanically advanced support pillars, and need to be receptive to movements due to the thermal expansions of the glazing layers [107]. Electrochromic windows are more widespread than thermochromic materials and therefore more economical. Their ability to adapt to external weather conditions demands precise and elaborate energy analysis and tuning, and the use of the appropriate chromogenic materials and control systems to achieve the maximum possible energy savings. These requirements set chromogenic window systems as complicated and high-priced solutions. In the future, chromogenic window systems will be a substantial component of the shift toward smart buildings equipped with suitable automation systems that will allow real-time adaptation to buildings' thermal energy loads and comfort demands.

Phase change materials demonstrate relatively low thermal resistance but high thermal capacity, as well as performance degradation over thermal cycles restricting their lifespan. PCMs are also characterized by a narrow temperature regulation range and hysteresis effect that demand precise compatibility analysis based on climatological conditions and building thermal needs [316]. Organic PCMs are less expensive than inorganic PCMs but do not present high fire resistance, which is an important prerequisite for building applications. The development of commercially available organic-based PCM composite materials and innovations regarding encapsulation techniques will be crucial for the development of a new kind of passive envelope technology characterized by a long lifespan and high thermal efficiency. Moreover, the high thermal mass passive solutions of the Trombe wall and solar chimney are reported to result in inconsistent heating during the winter period, and overheating during the summer period [173]. Therefore, these construction techniques must be combined with other technologies, for instance, thermal insulation, PCMs, cool coatings, or even ventilation systems, a fact that upgrades two simple passive solutions into multi-parameter complicated systems, ideal for integration into future smart buildings.

Cool-colored and daytime radiative materials offer only seasonal energy savings and cause overcooling deficiencies during winter periods. On the other hand, building-applied thermochromic coatings reduce overcooling during the heating period but offer restricted cooling savings in comparison to cool coatings. Additionally, the impact of thermochromic rooftop coatings on winter indoor thermal conditions is reported to deteriorate thermal comfort indicators by up to 170% [237]. The extensive and durable application of these technologies as a heat mitigation and energy-saving solution demands further research regarding optical photodegradation, aging, or photobleaching phenomena and further advancements in the domain of non-leuco dye thermochromic materials, characterized by higher optical stability, for instance, liquid crystals, photonic crystals, quantum dots, and plasmonic materials [187].

Mechanical ventilation with heat recovery can be an economical and energy-efficient solution only for airtight building constructions with low infiltration thermal losses, and therefore cannot be considered a standalone energy solution. Shading design needs to be included in building energy analysis because it can greatly impact a building's thermal

loads and indoor conditions. In contrast to the standard static shading component, dynamic shading has higher investment and maintenance costs due to its control system, making it an inappropriate solution for residential applications. Lastly, green infrastructure solutions demand high maintenance and appropriate water systems that can effectively regulate the soil's humidity and therefore offer important enhancements in the building's thermal performance.

## 8. Conclusions

In the present study, multiple energy-efficient building envelope techniques have been thoroughly reported and discussed regarding their energy-saving potential and impact on a building's thermal performance. Besides typical and conventional energy techniques, innovative materials and novel applications are the main focus of the study. The respective techniques regard insulation materials, window systems, techniques of high thermal mass property, optically advanced coatings with cooling potential, mechanical ventilation systems, and bioclimatic components and design concepts. The main conclusions of the present study concern the energy-saving potential in conjunction with the economic feasibility of the examined building envelope techniques and are given below:

- The incorporation of highly insulative, innovative solutions of vacuum-insulating panels and aerogel-based materials can result in important yearly energy savings (68.7% for the winter period and 30.0% for the summer period) but are affected by severe difficulties during the installation process because of their high brittleness and the risk of diminishing their thermal superiority in the case of perforation. High thermal performance can also be achieved with the selection of conventional, fire-resistant, and less expensive mineral wool and rock wool insulation materials. The future of insulating materials lies in the proper combination of their exceptional thermal properties, lightweight construction, minimum environmental footprint, and easy and fast installation. This trend is geared toward further research to minimize the risk of thermal degradation during the installation and to extend the lifespan of aerogels and vacuum insulation panels.
- Advanced, multi-glazed, gas-filled, solid insulation-filled, or vacuum window systems are heavy, highly efficient expensive constructions that can minimize radiation and convection thermal losses through openings of up to 55.0% (aerogel-filled) or even 76.0% (vacuum-filled). Window replacement combined with external thermal insulation addition belongs among the most economical and energy-saving optimum retrofit strategies for residences. An alternative but not commercially widespread solution, with satisfactory energy-saving potential, is electrochromic (18.5%) and thermochromic (50.0%) window systems that involve the use of appropriate control systems and chromogenic materials. Future window systems are smart, dynamic, and highly insulative windows that can properly adapt to environmental conditions and indoor thermal and optical comfort standards, integrated into building automation systems with real-time stimulus–response.
- Composite construction materials incorporating phase change materials are an innovative energy-efficient solution that secures high thermal mass exploitation, and easy and minimum-invasive application on the building envelope, protects the material from ambient conditions, and extends the phase change materials' lifespan. The induced decrease in cooling thermal load is reported to reach 54.0% for PCM-doped voided roof slabs. Future research should focus on the creation of organic-based PCM composite materials with expanded available temperature range options to fit the energy demand of any climatological conditions. Innovation in encapsulation techniques will be crucial for the extension of the lifespan of the materials and the enhancement of their thermal behavior and cost-effectiveness.
- Important thermal efficiency upgrades of the high thermal mass designs of the Trombe wall and solar chimney can be achieved through the incorporation of various components, for instance, thermal insulation, phase change materials (79.0%), highly



insulative window systems (12.2%), cool-colored materials (55.2%), or ventilation systems. This upgrade transforms two simple passive solutions into multi-parameter, dynamic systems, a suitable passive envelope energy-efficient solution to be integrated into future smart buildings.

- Cool-colored materials with static or dynamic optical properties applied on a building's external envelope in the form of coatings can induce important seasonal (17.0% cooling savings with cool-colored materials of static properties) or annual energy savings (19.0% annual savings with cool-colored materials of dynamic properties). Despite the domain's important progression, further research is demanded for the abatement of optical degradation and success of optical stability and greater lifespan of these materials.
- The enhancement of a building's thermal performance and indoor thermal comfort can be achieved through the incorporation of mechanical ventilation systems with heat recovery (instantaneous load energy savings of up to 50.0%), green façades and green walls (annual energy savings of 45.0%), shading components (annual energy savings of 50.4%), and cool-colored materials. Parallely, the incorporation of these passive envelope technologies will guarantee more energy-sufficient buildings, while their incorporation in a city's wider building infrastructure, for instance, pavements or open public spaces, will result in the heat mitigation and alleviation of heat extremes and peak city-level air temperatures.

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## Nomenclature

|   |                                                                               |
|---|-------------------------------------------------------------------------------|
| R | Thermal resistance, $\text{m}^2 \cdot \text{K} / \text{W}$                    |
| t | Thickness, m                                                                  |
| U | Thermal conductance (transmittance), $\text{W} / (\text{m}^2 \cdot \text{K})$ |

## Superscripts and Subscripts

Cond    Conduction

## Greek Symbols

$\lambda$     Thermal conductivity,  $\text{W} / (\text{m} \cdot \text{K})$

## Abbreviations

|      |                                  |
|------|----------------------------------|
| DS   | Double sash                      |
| EU   | European Union                   |
| EPS  | Expanded Polystyrene             |
| MVS  | Mechanical ventilation system    |
| PCM  | Phase change materials           |
| PIR  | Polyisocyanurate                 |
| PU   | Polyurethane                     |
| PVC  | Polyvinyl chloride               |
| SS   | Single sash                      |
| uPVC | unplasticized polyvinyl chloride |
| VIP  | Vacuum-insulating panel          |
| XPS  | Extruded Polystyrene             |

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